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Simultaneous Design and Control of Energy Systems

A thesis submitted to attain the degree of

DOCTOR OF SCIENCES of ETH ZURICH

(Dr. sc. ETH Zurich)

presented by

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To my family

Acknowledgments

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I am eternally thankful to my family, my parents, Masoud and Nezhat, my sister, Arghavan, and my beloved wife, Sahar. Without their love and understanding I would not have made it this far.

Last but not least, I would like to thank my friends for their unconditional support and kindness.

Araz Ashouri
April 2014

Two roads diverged in a wood, and I–
I took the one less traveled by,
And that has made all the difference.

— *Robert Frost*

Abstract

The comprehensive design of building energy systems incorporates the tasks of selection, dimensioning and control of devices. A simultaneous acquirement of these tasks is a necessity to achieve an overall optimal design. However, such mutual optimizations become a complex problem, implying a high computational effort. On the other hand, the concept of Smart Buildings is being introduced as a result of the altered boundary conditions such as directives, sustainability considerations, energy tariffs, user demands, and the emerging novel energy supply which is called the Smart Grid. Furthermore, due to the stringent requirements for integration of renewable energy sources and storage systems in future smart building systems, their complexity of design and control will increase inevitably.

The goal of this thesis is to develop a design framework for the optimal selection, dimensioning, and control of smart building systems, in order to investigate the potentials and tackle the problems of such comprehensive design. In this framework, namely the smart building designer, various building services such as thermal and electrical storages, heating and cooling systems, and renewable energy sources are modeled and implemented using mixed-integer linear programming techniques. In order to enable a reasonable comparison of various configurations, optimal operating strategies are computed in parallel. The implementation as a mixed integer linear programming problem allows the evaluation of the optimal system operation for an entire year in less than a minute on an average desktop computer. As a result, analyses on the sensitivity of designs to various boundary conditions such as governmental legislation and ambient conditions are carried out within reasonable amount of time.

Using the design framework, several case studies are conducted. The first case study deals with the consequences of variable energy tariffs,

while the second one is concerned with the limitation of external energy demand, similar to the "Zero Energy Building" initiative. The third case study considers the usage of storages. Since the building cooling systems are responsible for high peak loads and because their associated costs of operation account for a considerable share of the total lifetime costs, specific attention is paid to design and operation of the cooling system including the vapor compression system, the hybrid dry cooler, and the ice storage unit. It is concluded that the system including an optimally designed ice storage achieves lifetime cost reductions of approximately 8% by reducing the operational costs and the investment costs for the downsized refrigeration machine. Also, the impact of two-level and variable pricing systems on the design of the cooling system are examined.

An extension of the framework provides a robust design against various uncertainties in the design phase. A great challenge to the design problem appears once the uncertainties of boundary conditions such as weather conditions, user demands and energy costs are taken into account. An approach to protect the suggested system configuration against the possible uncertainties is a stochastic optimization which results in a robust design. Therefore, the resulting designs of the deterministic and stochastic optimizations are compared for several uncertainties. The comprehensive analyses show that the achievement of a robust design is computationally demanding and not even desirable in general. However, the size of devices may vary by up to 100% when a robust design is attained.

Zusammenfassung

Die umfassende Auslegung von Gebäudeenergiesystemen beinhaltet die Auswahl, die Dimensionierung und die Betriebsstrategie der Gebäudetechnikelementen. Für eine optimale Auslegung müssen diese drei Elemente in einem Schritt bearbeitet werden, wobei diese gleichzeitige Optimierung eine komplexe Aufgabenstellung ist, verbunden mit einem hohen Rechenaufwand.

Zusätzlich werden die Rahmenbedingungen für den Betrieb von Gebäuden immer anspruchsvoller. Sogenannte „Smart Buildings“ wurden eingeführt, um diesen geänderten Rahmenbedingungen wie gesetzliche Vorschriften, Nachhaltigkeitsüberlegungen, neue variable Energietarife, Ansprüche von Benutzern und der zunehmenden Verbreitung von erneuerbaren Energiequellen auf Gebäudeseite gerecht zu werden. Der Begriff von intelligenten Elektrizitätsnetzen sogenannte „Smart Grids“ sind auf der Vorsorgungsseite schon länger eingeführt worden. Infolge der geänderten Rahmenbedingungen ist der Einsatz von Speichersystemen und erneuerbaren Energiequellen in Gebäudesystemen interessant geworden, verbunden mit einer erhöhten Komplexität des Systems und dessen Auslegung und Betriebsstrategie.

Das Ziel dieser Arbeit ist es, eine Grundstruktur für die integrale Auslegung von Gebäudeenergiesystemen zu schaffen, mit der die Potentiale von solchen Systemen analysiert werden kann. Der sogenannte „Smart Building Designer“ beinhaltet diverse gebäudetechnische Elemente wie thermische und elektrische Speicher, Wärme- und Kälteerzeugungsanlagen und erneuerbare Energieressourcen. Diese Komponenten sind als ganzzahlig lineares Programmierungsproblem modelliert. Um verschiedenste Topologien und Fragestellungen beantworten zu können, werden die Betriebstrategien der Systeme gleichzeitig hergeleitet. Infolge der Modellierung als ganzzahlig lineares Problem, können die Betriebstrategien für ein ganzes Jahr in wengiger als einer Minute auf einem norma-

len Desktopcomputer gelöst werden. Dadurch können die Sensitiväten des Gebäudesystems auf unterschiedlichste Randbedingungen innerhalb nützlicher Frist analysiert werden.

Mittels dieses Frameworks wurden diverse Fallstudien durchgeführt. In der Ersten wurden optimale Energiesysteme unter variablen Energiepreisen analysiert. In der Zweiten wurden die Konsequenzen einer Nullenergie-Gebäude Gesetzgebung geprüft. In der Dritten wurde im Detail der Einsatz von Speichern in Gebäuden untersucht, um Spitzenbezüge aus dem elektrischen Netz zu limitieren und damit die Leistungskosten zu minimieren, welche einen Grossteil der Elektrizitätskosten ausmachen können.

Im speziellen wurde der Einsatz und Auslegung von Eisspeichern in Kühlsystemen untersucht. Die Lebenszykluskosten können mit deren Einsatz um ungefähr acht Prozent gesenkt werden, wobei ein Teil der Einsparung durch eine Reduktion der Betriebskosten und ein Teil der Einsparung durch kleiner dimensionierte Kältemaschinen erzielt wird. Ebenfalls untersucht wurde die unterschiedliche Dimensionierung des Kältesystems unter diversen Elektrizitätstarifsystemen.

In einem zweiten Schritt wurde die Grundstruktur erweitert, um Unsicherheiten in der Auslegungsphase bei der Optimierung zu berücksichtigen und so eine robuste Lösung zu erhalten. Eine grosse Herausforderung für das Dimensionierungsproblem sind die Einführung von Unsicherheiten betreffend den Aussenbedingungen, dem Nutzerverhalten und den zukünftigen Energiepreisen. Mittels einer stochastischen Optimierung kann die Auslegung des Gebäudesystems robuster gegen diese Unsicherheiten gemacht werden. Um die Robustheit zu beurteilen sind intensive Untersuchungen durchgeführt worden, welche gezeigt haben, dass die Lösungen des stochastischen Optimierungsproblem deutlich rechenintensiver sind und nicht immer zu merklich besseren Resultaten führen. Dennoch kann die Auslegung der einzelnen Gebäudetechnikkomponenten im Falle einer robusten Dimensionierung zum Teil bis zu hundert Prozent unterschiedlich sein, verglichen mit der deterministischen Lösung.

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Preface

As mandated by several directives issued by the European Union, starting from 2020, new buildings must be designed and constructed as low-energy buildings [1]. This situation is to be achieved mostly by an installation of renewable energy technologies, storage systems, and improved insulations. As a result, the capital cost of these building systems and thus the potential for optimization will grow significantly. In addition, on the electric supply side, a smart grid with time-varying electricity tariffs may be established. These two directions of impact offer the opportunity to optimize the interactions between the so-called smart building and the distribution grid.

Buildings and Energy Sector

The building infrastructure plays a central role in modern society and its energy demand, since about 40% of the energy utilization occurs within the building sector [2]. To be more precise, the share of energy consumption for residential and commercial buildings is around 49% in Switzerland and 41% in the United States [3, 4]. Usually, a large part of this energy is used for space heating or cooling and the ventilation system [4], implying the necessity for the optimal operation of such services. However, reducing the energy utilization is not the only goal and there are more issues to be considered.

First of all, it is expected that in the foreseeable future the supply of oil and gas will not be able to satisfy the rising demand, which will lead to a massive increase of fossil fuel prices [5]. This expectation motivates the urge for seeking alternative solutions and more efficient ways of supplying heating, cooling and electricity demand.

Concurrently, the legal regulations for the use of fossil fuels and CO_2 emissions of buildings will be further restricted [1] resulting in the introduction of more decentralized production facilities such as building-integrated photovoltaic systems. Such regulations should also be considered during the design phase of the building.

Finally, with the introduction of a "smart grid" on the supply side, the impact of the interactions between the building and the supply network increases. In addition, introducing renewable energy sources such as wind and solar power plants will result in a fluctuating electricity production, raising a significant need for balancing power in the electric grid. Therefore, the so-called "smart building" should be able to handle real-time interactions with the grid.

The process of design and control of such a smart building system and its services is a computationally complex task. The variety of building applications, local weather conditions, governmental restrictions and energy tariffs makes it more difficult to generalize the process of optimal component design. Therefore, the goal of this thesis is to develop a fast and modular framework which optimizes the design and control of such complex energy systems.

1 Introduction

Figure 1.1 depicts a schematic of typical steps in the process of a building system design. Usually, the boundary conditions such as weather parameters, usage specifications and pricing for energy carriers are considered in the first step. Also, in many countries there are governmental directives which may apply limitations on the consumption of energy, CO_2 emissions, etc. The second and the most challenging step is the optimal selection and dimensioning of building services, as well as finding an efficient operation strategy. Finally, an evaluations of the operational and discomfort costs gives an insight into the capability of the design and provides a measure to determine the optimality of the system.

1.1 Simultaneous Design and Control

The performance of building systems highly depends on the selection and dimensioning of components and devices. The variety of building applications, local weather conditions, governmental legislation and energy prices makes it difficult to generalize the process of an optimal design. Therefore, it is useful to formulate the design problem as a model-based optimization which includes the most relevant dynamics of the building system.

In addition to the selection and dimensioning of devices, the efficient operation of the building services is essential. Especially, if storage devices or limited power resources are present, sophisticated energy management systems are required. This task turns out to be very challenging, since the optimal design and control are not two independent problems and must be performed simultaneously to address the full potential [6]. Hence, the design task becomes even more sophisticated.

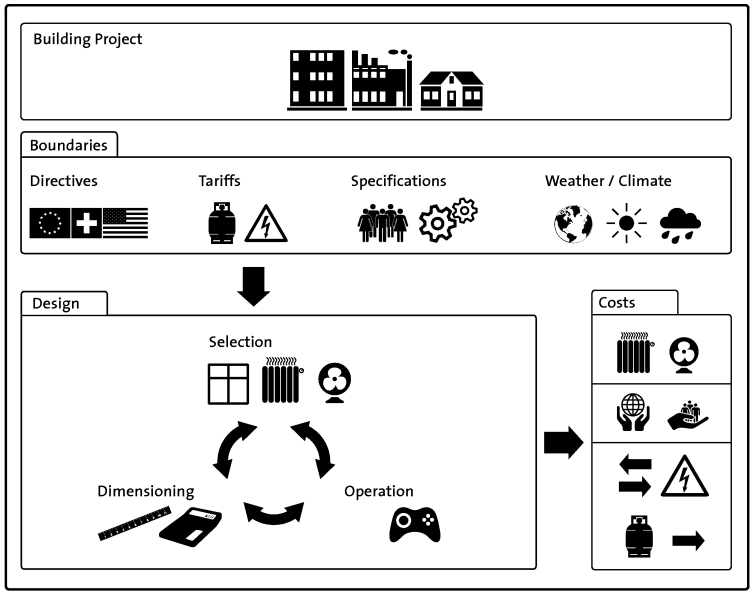


Figure 1.1: Overview of the regular steps in a building project: (1) Defining the boundary conditions; (2) Simultaneous optimization of selection, dimensioning and operation of devices; (3) Evaluations of the operational and discomfort costs.

1.2 State of the Art and Motivation

Many researchers have already targeted specific problems of building design. For example, in [7, 8] the authors have considered the thermal subsystems and in [9–11] the electric subsystems. Others have considered a combination of both subsystems, but within a limited time period [12] or without considering energy storage devices [13]. Also, when several building components are optimized for long periods, many authors have limited the problem to specific cases and do not provide a general framework [14–17]. In this thesis, the development of a comprehensive design framework for a total building optimization is presented. The goal is to cover the complete building system, including the most relevant devices and boundary conditions, and to provide a simultaneous optimal system design and control. Furthermore, the unavoidable uncertainties in the

boundary conditions are studied and their effects on the final design are investigated.

1.3 Structure of the Thesis

In Chapter 2 the main structure of the design framework as well as the devices modeled is provided. The first part of Chapter 3 is dedicated to the formulation of the deterministic design. In the remainder of the chapter, two case studies are shown. The first case study includes the general optimization of a commercial building, plus several investigations on the effect of global constraints on the final design. The second case study provides a deeper insight into the optimization of cooling systems, the advantages of storage systems, and the interaction with the power grid. The first part of Chapter 4 describes the most common uncertainties on the boundary conditions, leading to a necessity for the introduction of the robust design concept. A case study follows in the rest of the chapter which brings about the opportunity for a comparison of deterministic and stochastic approaches. Finally, the main results and discussions are shown in Chapter 5 which concludes the text.

1.4 Main Publications

The following scientific publications are a result of this research:

- A. Ashouri, F. Petrini, R. Bornatico, and M.J. Benz. Sensitivity Analysis for Robust Design of Building Energy Systems. Accepted for publication in *Energy*, August 2014.

- A. Ashouri, S.F. Fux, M.J. Benz, and L. Guzzella. Optimal design and operation of building services using mixed-integer linear programming techniques. *Energy* 2013, Vol. 59, Pages 365–376.

- S.F. Fux, A. Ashouri, M.J. Benz, and L. Guzzella. EKF based self-adaptive thermal model for a passive house. *Energy and Buildings* 2014, Vol. 68, Pages 811–817. (Accepted for publication in 2012.)

- A. Ashouri, M.J. Benz, R. Stettler, S.F. Fux, and L. Guzzella. Heating and cooling demand estimation using a self-learning thermal building model. *Proceedings of the CISBAT Conference*, Lausanne, Switzerland, September 2011.

- M.J. Benz, A. Ashouri. Smart Building Designer: Eine Methode zur optimalen Auslegung von komplexen Gebäudesystemen. *Status-Seminar an der ETH Zürich: Brenet*, Switzerland, September 2012.

- A. Ashouri, M.J. Benz, L. Guzzella. Comparing control strategies for EV and PHEV fleets providing regulation ancillary services. *Proceedings of the IEEE International Conference on Control Applications (CCA)*, Croatia, October 2012.

- S.F. Fux, M.J. Benz, A. Ashouri, L. Guzzella. Short-term thermal and electric load forecasting in buildings. *Proceedings of the CISBAT Conference*, Lausanne, Switzerland, September 2013.

2 Modeling and Problem Formulation

The model framework is presented in this section. A set of the most relevant building services are selected to be included in the optimal design process. Building services are described by means of static or first-order linear differential equations, based on the basic physical laws.

2.1 MILP

The behaviors of devices are modeled using the mixed-integer linear programming (MILP) technique, which is an appropriate method for the formulation of optimization problems [18]. This technique has been used by many researchers to perform energy optimization for building services [19–21], while some have chosen quadratic optimization at a cost of not reaching an absolute optimal point [22]. An advantage of utilizing MILP is the availability of powerful solver packages which are either non-commercial (e.g., SCIP [23] and CBC [24]) or free for academic use (e.g., CPLEX [25] and Gurobi [26]).

2.2 Model Framework

One challenging aspect of modeling a building system is the commonly large number of interconnections among the subsystems, making it cumbersome to keep track of all interactions. Therefore, the devices with a similar behavior are categorized into the same subsystems and the energy flows with the same destination or origin are merged. Such an arrangement provides a maximal modularity for the SBD framework.

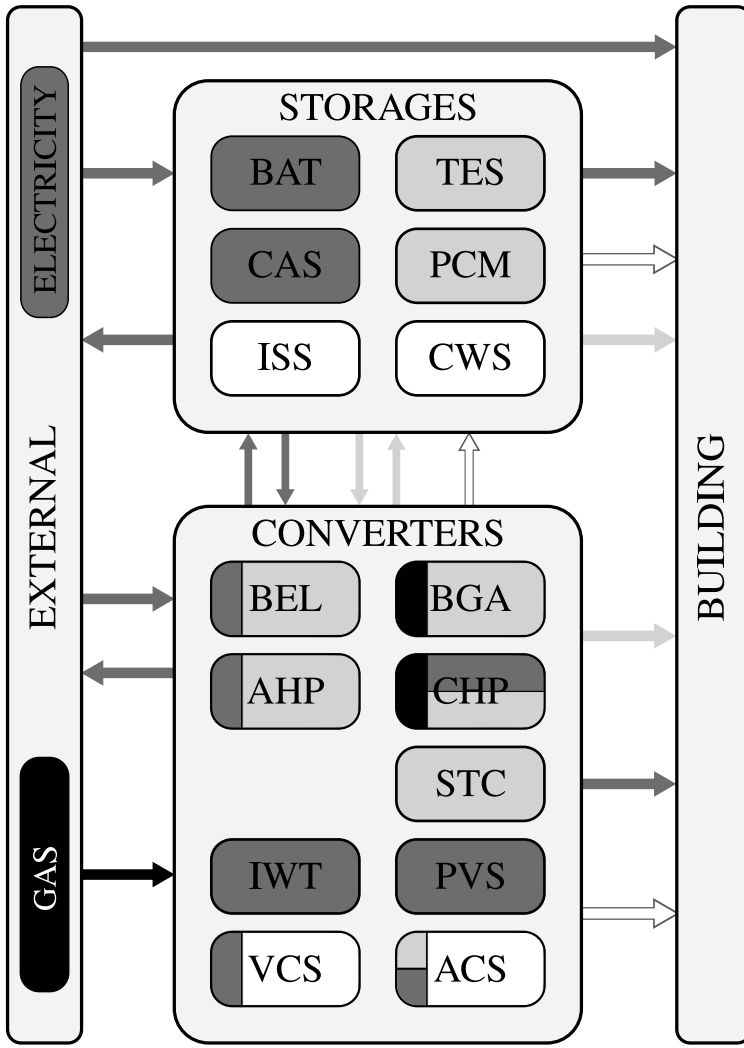


Figure 2.1: Overview of the SBD blocks, subsystems and energy flows: gas flow (black), electricity flow (dark grey), heating flow (light grey) and cooling flow (white).

The model framework is defined as follows: Every building device is considered to be either a converter or a storage of a type of energy.

Also, there is a mutual dependency between the electrical, chemical and thermal infrastructures [27]. This setup is shown in Figure 2.1. Energy flows between the external sources (EXT), storage block (STO), converter block (CON), and the building (BDG) are indicated with arrows. Flows of gas, electricity, heating and cooling are separated using different colors. For storages, the color represents the type of stored energy. For converters, single color shading indicates the type of output energy. If more than one color is used, the colors to the left and right indicate the type of input and output energy, respectively. For example, the combined heat and power (CHP) device converts chemical energy (black) into electricity (dark grey) and heat (light grey). Clearly, the internal power split within each of these blocks is not a trivial problem. Such detailed energy transactions are obtained by the SBD framework as the solution of an optimization problem.

2.2.1 Formulating Conventions

The SBD framework is implemented in MATLAB, and the modeling toolbox Yalmip [28] is used as an interface to the MILP solver. The following naming convention is used in the rest of the thesis:

- An *italic* letter represents a scalar variable, while a **bold roman** letter stands for a vector. As an example, the symbols E and $\mathbf{E}$ stand for the electric energy in scalar and vector format, respectively. These are derived from the electric power vector $\dot{\mathbf{E}}$ using:

$$\mathbf{E}_{\mathrm{XYZ}\rightarrow\mathrm{ZYX}} = \dot{\mathbf{E}}_{\mathrm{XYZ}\rightarrow\mathrm{ZYX}} \cdot t_{\mathrm{s}} \tag{2.1a}$$

$$E_{\mathrm{XYZ}\rightarrow\mathrm{ZYX}} = \sum_{j=1}^{N} \mathbf{E}_{\mathrm{XYZ}\rightarrow\mathrm{ZYX}}(j) \tag{2.1b}$$

  where XYZ and ZYX are two arbitrary devices or blocks, t_{s} is the sampling time and j represents the time index, up to the optimization duration N. Scalar and vector annotation are mainly used to distinguish between the design of devices (one single value over the whole optimization period) and optimal control of devices (a vector with one value per sample time).

- A superscript d symbolizes an independent decision variable. As a result, $\dot{\mathbf{Q}}^d$ would be a controlled thermal power manipulated by the optimizer.

- The zero-power option is implemented to indicate the presence of a certain device. The binary variable χ_{XYZ} takes a value of $'1'$ if the device XYZ is included in the building design and $'0'$ if the device should be not considered in the current investigation, respectively.

The following sections will describe the structure of the three main blocks in the model, namely the building, the converter block and the storage block.

2.2.2 General Assumptions and Simplifications

Several simplifications are considered in the process of developing the framework. To begin with, the time step is defined as one hour. Although this will lead to a decreased accuracy for describing the system dynamics, it allows for longer optimization periods such as one year. Also, during a time step the inputs are kept constant, similar to a zero order hold functionality. This implementation also simplifies the process of discretizing the physical models.

Furthermore, parameters of the model which depend on other uncontrolled variables or external inputs are pre-calculated before the optimization starts. This is mainly to avoid a nonlinear optimization problem. As an example, the coefficient of performance (COP) of the air-source heat pump is a function of ambient temperature; hence it is pre-calculated using the weather data of a typical meteorological year.

2.3 Building Block

The building block represents the overall behavior of the building envelope and integrates several static and dynamic properties. The model mainly consists of the following components:

Thermal Behavior

The thermal behavior of the building which defines the thermal performance of structures is associated with various physical and design parameters such as the heat capacity, thermal transmittance and also the space distribution [29]. More details on the thermal model implemented in SBD is provided later in this Chapter.

Ventilation System and Infiltration

The ventilation system guarantees the fresh air recirculation, while the air infiltration rate is a function of the number of occupants. The SBD includes an optimal heat recuperation from the exhaust air as well, resulting in an improved efficiency for the ventilation system. The concept of free cooling is also implemented which allows a direct or indirect inflow of cooling power from the ambient air during the night-time. In case of an indirect inflow, the cooling power is stored in order to be used later for cooling the building.

Blind System and Smart Lighting

The optional usage of automated blind system and smart lighting is implemented in SBD. Blinds are opened and closed optimally to allow enough light to pass through the windows, as well as regulating the heat flow as a result of solar irradiations. The smart lighting system guarantees an adequate light for the occupants, based on the parameters such as the application of the building, the instantaneous electricity price, and the current state of the blinds.

Discomfort Concept

In order the maintain the comfort for the occupants at all times, the deviation of the building temperature from a set-point value is converted to an equivalent monetary value, using the concept of discomfort costs. More detailed information regarding the implementation in SBD as well as an example are provided in Chapter 3.

2.3.1 Building Thermal Model

Various thermal modeling methodologies [30] and numerous dedicated tools have been developed for the simulation of building thermal behavior (for an overview see e.g. [31]). Due to their ability to describe the building thermal behavior accurately despite their simplicity, lumped-parameter models based on basic physical principles are being used frequently [32].

2.3.2 Resistance-Capacitance Networks

Typically, lumped-parameter thermal building models are represented by resistance-capacitance networks. In this electric circuit analogy, an electric capacitance represents the thermal capacity of a wall layer, for instance, or the indoor air and the corresponding voltage represents the temperature. The heat flow is equivalent to the electric current, such that an electric resistance represents a thermal resistance.

Several publications have described applications of lumped-parameter thermal building models on buildings with passive solar features. Park et al. [33] analyzed a 1R1C model identifying the model parameters using measurement data from a well-insulated test room where four lamps were the only heat source. The reported data shows good agreement between the simulated results and the measurement data.

2.3.3 The 1R1C Model

For the purpose of this investigation, all relevant rooms are lumped into only one thermal zone. Figure 2.2 shows the electric analogy of the 1R1C lumped-parameter thermal building model.

The 1R1C model is the simplest dynamic lumped-parameter thermal building model with one resistance and one capacitance. In this model, the ambient air temperature is shown by T_{amb}. The inverse thermal transmittance of the outside walls and windows is represented by the resistance $^{1}/_{\mathrm{U_{BDG}}}$, while thermal capacities of the room air, furniture and internal constructions are lumped into one effective capacitance C_{BDG} with the average building temperature T_{BDG}. The heat flows acting on the capacitance are integrated into the single variable $\dot{Q}_{\mathrm{BDG,in}}$,

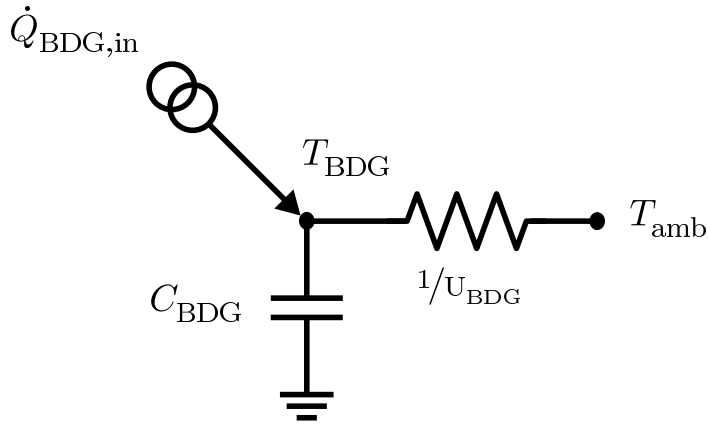


Figure 2.2: Electrical analogy of the lumped-parameter 1R1C building thermal model.

which includes any solar heat gains through windows $\dot{\mathrm{Q}}^{\mathrm{sol}}_{\mathrm{BDG}}$, the heating load $\dot{\mathrm{Q}}_{\mathrm{BDG}}$, the cooling load $\dot{\mathrm{Q}}'_{\mathrm{BDG}}$, the heat gain of electric equipments $\dot{\mathrm{E}}_{\mathrm{BDG,eqp}}$, and the heat gain caused by the occupants $\dot{\mathrm{Q}}^{\mathrm{occ}}_{\mathrm{BDG}}$.

As a result, the energy balance of the building is described by the following differential equation:

$$C_{\mathrm{BDG}} \cdot \frac{d}{dt} T_{\mathrm{BDG}} = U_{\mathrm{BDG}} \cdot (T_{\mathrm{amb}} - T_{\mathrm{BDG}}) + \dot{\mathrm{E}}_{\mathrm{BDG,eqp}} + \dot{\mathrm{Q}}^{\mathrm{occ}}_{\mathrm{BDG}} + \dot{\mathrm{Q}}^{\mathrm{sol}}_{\mathrm{BDG}} + \dot{\mathrm{Q}}_{\mathrm{BDG}} - \dot{\mathrm{Q}}'_{\mathrm{BDG}} \quad (2.2)$$

The solar heat gains through windows depend on the total solar radiation on the windows and the angle-dependent total solar energy transmittance of the windows. In this work, the Perez model described in [34] is used to compute the beam, diffuse and ground reflected solar radiation on the sloped windows from the horizontally measured global and diffuse radiation data. The angle dependency of the total solar energy transmittance is evaluated using the relationship proposed by Karlsson and Roos [35].

2.3.4 MILP Modeling of the Building Block

The thermal behavior of the building is modeled using a lumped parameter resistance-capacitance approach.

For our purposes the single-state 1R1C thermal model is chosen, since it is shown by authors in [36] that such a model provides a reasonable compromise between simplicity and accuracy. However, more complex representations of the building thermal behavior are possible, as long as the model remains control-oriented and linear. The enthusiastic reader is referred to [36] for more discussion on the subject. The MILP formulation of the building block is shown in Eq. (2.3). The equation consists of constraints imposed on the decision variables (Eq. (2.3a–2.3f)) as well as the state equation (Eq. (2.3g)) which updates the building temperature vector $\mathbf{T}^{\mathrm{d}}_{\mathrm{BDG}}$.

The temperature is initiated at the set-point (Eq. (2.3a)) and is kept within the acceptable boundaries to maintain the user comfort (Eq. (2.3b)). Since there exist vector and scalar variables, an index i is used to indicate the ith element of a vector, in order to avoid unclearness in mathematical operations. The electrical connections among the building and other blocks are presented by Eq. (2.3c). The available heating and cooling power are calculated in Eq. (2.3d) and (2.3e), respectively. The parameters η_{BDG} and η'_{BDG} account for transmission losses from the pipes or ducts, for heating or for cooling. Respectively, there is no need to consider a separate transmission efficiency for heating/cooling devices. In Eq. (2.3d), the demand for the domestic hot water (dhw) is also considered. In Eq. (2.3f) the inlet thermal power is calculated as a sum of the electric equipment including lighting (eqp), occupants (occ), and solar (sol) heat gains, plus the heating or cooling power provided by other blocks in the building.

$$\mathbf{T}^{\mathrm{d}}_{\mathrm{BDG}}(1) = T^{\mathrm{set}}_{\mathrm{BDG}} \tag{2.3a}$$

$$T^{\mathrm{min}}_{\mathrm{BDG}} \leq \mathbf{T}^{\mathrm{d}}_{\mathrm{BDG}}(i) \leq T^{\mathrm{max}}_{\mathrm{BDG}} \tag{2.3b}$$

$$\dot{\mathbf{E}}_{\mathrm{BDG,eqp}} = \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{EXT}\rightarrow\mathrm{BDG}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\rightarrow\mathrm{BDG}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{STO}\rightarrow\mathrm{BDG}} \tag{2.3c}$$

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BDG}} \leq \eta_{\mathrm{BDG}} \cdot \left(\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{CON}\rightarrow\mathrm{BDG}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{STO}\rightarrow\mathrm{BDG}} \right) - \dot{\mathbf{Q}}^{\mathrm{dhw}}_{\mathrm{BDG}} \tag{2.3d}$$

$$\dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{BDG}} = \eta'_{\mathrm{BDG}} \cdot \left(\dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{CON}\rightarrow\mathrm{BDG}} + \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{STO}\rightarrow\mathrm{BDG}} \right) \tag{2.3e}$$

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BDG,in}} = \dot{\mathbf{E}}_{\mathrm{BDG,eqp}} + \dot{\mathbf{Q}}^{\mathrm{occ}}_{\mathrm{BDG}} + \dot{\mathbf{Q}}^{\mathrm{sol}}_{\mathrm{BDG}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BDG}} - \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{BDG}} \tag{2.3f}$$

$$\begin{aligned} \mathbf{T}^{\mathrm{d}}_{\mathrm{BDG}}(i) = & \exp\left(\frac{-t_{\mathrm{s}} \cdot U_{\mathrm{BDG}}}{C_{\mathrm{BDG}}}\right) \cdot \mathbf{T}^{\mathrm{d}}_{\mathrm{BDG}}(i-1) \\ & + \left(1 - \exp\left(\frac{-t_{\mathrm{s}} \cdot U_{\mathrm{BDG}}}{C_{\mathrm{BDG}}}\right)\right) \cdot \mathbf{T}_{\mathrm{amb}}(i) \\ & + \frac{1}{U_{\mathrm{BDG}}} \cdot \left(1 - \exp\left(\frac{-t_{\mathrm{s}} \cdot U_{\mathrm{BDG}}}{C_{\mathrm{BDG}}}\right)\right) \cdot \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BDG,in}}(i) \ , \\ & , \ i = 2, 3, ..., N. \end{aligned} \tag{2.3g}$$

The Eq. (2.3g) is a discrete-time version of the Eq. (2.2). Here, C_{BDG} represents the thermal capacity and t_{s} is the time step used for the discretization of the differential equation. The thermal transmittance U_{BDG} is calculated as:

$$U_{\mathrm{BDG}} = \frac{1}{R_{\mathrm{BDG,amb}}} + \frac{1}{R_{\mathrm{BDG,ven}}} \tag{2.4}$$

where $R_{\mathrm{BDG,amb}}$ stands for the thermal resistance between the building envelope and the ambient air through the walls and windows, while $R_{\mathrm{BDG,ven}}$ represents the thermal resistance due to ventilation and infiltration of air. Some nominal values are taken from the building design guidelines of the Swiss Architects Association [37].

2.4 Converter Block

Certain devices are installed in the building in order to convert the available resources into the appropriate deliverable types of energy. The outputs of the converters are transferred to the building envelope in the form of heating or cooling power, or electricity. However, the energy management system or the optimizer (in the design phase) decides how the power must flow among the converter devices and other parts of the system. As an example, the air source heat pump (AHP) consumes elec-

tricity and provides power for area heating or domestic hot water. The electricity input is provided to AHP either directly from the external grid ($\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{EXT}\to\mathrm{CON}}$), from an electricity storage ($\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{CON}}$), or even from another converter device, such as a photovoltaic system ($\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{CON}}$). Also, the output of AHP may immediately be transferred to the building, or be dispatched to a storage or another converter device. Figure 2.3

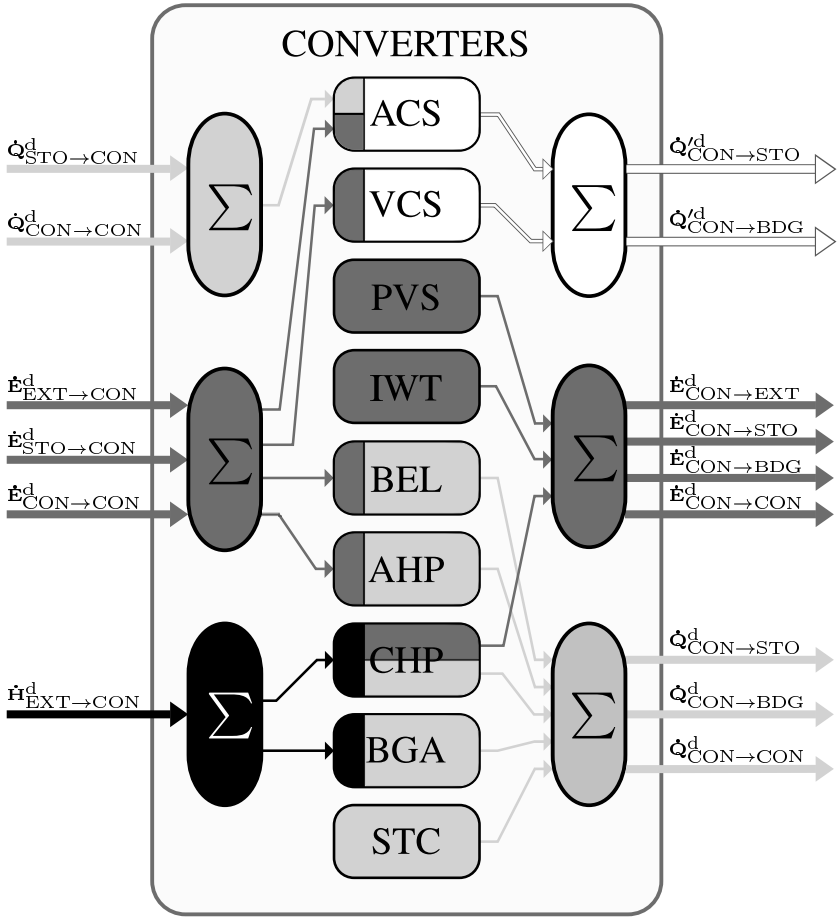


Figure 2.3: Overview of the external and internal connections of the converter block: gas flow (black), electricity flow (dark grey), heating flow (light grey) and cooling flow (white).

shows the interactions described. In this setup, the optimizer controls the total input and output energy flows, as well as the internal flows between any two converter devices. The input and output power flows of each converter are still subject to the optimization step, although they are not implemented as independent control variables. For example, the electricity flows in the converter block are described as:

$$\begin{aligned}\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{EXT}\to\mathrm{CON}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{CON}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{CON}} \\ = \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{ACS,in}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{VCS,in}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{AHP,in}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{BEL,in}}\end{aligned} \tag{2.5a}$$

$$\begin{aligned}\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{EXT}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{STO}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{BDG}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{CON}} \\ = \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{PVS,out}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{IWT,out}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CHP,out}}\ .\end{aligned} \tag{2.5b}$$

The variables on the left-hand side of Eq. (2.5) help certain constraints to be defined more clearly. If the output of a convertor is beyond the actual demand, the extra energy can be used by another converter to produce a more desired type of energy. This phenomenon become more interesting if renewable energy sources are available, hence efficiency is not a concern. For this reason, it is possible in SBD to transfer the energy between converters (CON → CON), which is in fact an internal feedback. The following subsections present a short description of every converter device that is modeled and that is included in the SBD framework.

2.4.1 Photovoltaic System

A simple, yet accurate and fast control-oriented linear model is used for the photovoltaic system (PVS). Eq. (2.6b) shows the calculation of the electric power output of the PVS using its area $A^{\mathrm{d}}_{\mathrm{PVS}}$, its efficiency vector $\boldsymbol{\eta}_{\mathrm{PVS}}$, and the vector of total solar incident on the PVS surface $\mathbf{I}_{\mathrm{PVS}}$, as suggested by [34].

$$\chi^{\mathrm{d}}_{\mathrm{PVS}} \cdot A^{\mathrm{min}}_{\mathrm{PVS}} \leq A^{\mathrm{d}}_{\mathrm{PVS}} \leq \chi^{\mathrm{d}}_{\mathrm{PVS}} \cdot A^{\mathrm{max}}_{\mathrm{PVS}} \tag{2.6a}$$

$$\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{PVS,out}}(i) = A^{\mathrm{d}}_{\mathrm{PVS}} \cdot \boldsymbol{\eta}_{\mathrm{PVS}}(i) \cdot \mathbf{I}_{\mathrm{PVS}}(i) \tag{2.6b}$$

$$\mathbf{T}_{\mathrm{PVS}}(i) = \frac{U_{\mathrm{PVS}} \cdot \mathbf{T}_{\mathrm{amb}}(i)}{U_{\mathrm{PVS}} - \pi_{\mathrm{PVS}} \cdot \mathbf{I}_{\mathrm{PVS}}(i)} + \frac{\mathbf{I}_{\mathrm{PVS}}(i) \cdot \left(f_{\mathrm{PVS}} - \eta_{\mathrm{PVS,ref}} - \pi_{\mathrm{PVS}} \cdot T_{\mathrm{PVS,ref}} \right)}{U_{\mathrm{PVS}} - \pi_{\mathrm{PVS}} \cdot \mathbf{I}_{\mathrm{PVS}}(i)} \tag{2.6c}$$

$$\boldsymbol{\eta}_{\mathrm{PVS}}(i) = \eta_{\mathrm{PVS,ref}} - \pi_{\mathrm{PVS}} \cdot \left(\mathbf{T}_{\mathrm{PVS}}(i) - T_{\mathrm{PVS,ref}} \right) \tag{2.6d}$$

$$A_{\mathrm{PVS}}^{\mathrm{max}} = l_{\mathrm{BDG}} \cdot w_{\mathrm{BDG}} \ . \tag{2.6e}$$

According to [38], the efficiency of the PVS is affected by the module temperature T_{PVS}, which is precalculated using Eq. (2.6c). The thermal transmission coefficient U_{PVS} and the temperature coefficient of efficiency π_{PVS} depend on the type of the PVS and are provided by the manufacturer. The portion of the solar incident passing through the PVS glass is denoted by the factor f_{PVS}, while $T_{\mathrm{PVS,ref}}$ and $\eta_{\mathrm{PVS,ref}}$ are the reference temperature and efficiency, respectively.

Eq. (2.6a) applies the upper and lower boundaries on the surface area of the PVS, which is a design parameter. If the PVS is not included in the design process, the existence variable $\chi_{\mathrm{PVS}}^{\mathrm{d}}$ forces the area to be zero. A theoretical limitation on the $A_{\mathrm{PVS}}^{\mathrm{max}}$ would be the area of the roof, calculated using the length l and the width w of the building (Eq. (2.6e)). However, normally there are some obstacles on the roof making a part of it not usable for installing a PVS. Also, the PVS modules are usually mounted with an inclination, hence the calculations depend on the geometries of the roof.

2.4.2 Solar Thermal Collector

The solar thermal collector (STC) absorbs sunlight and converts it to heat. The amount of solar power absorbed depends on the STC surface area A_{STC}, the total solar incident on the STC surface $\mathbf{I}_{\mathrm{STC}}$, and the ambient temperature $\mathbf{T}_{\mathrm{amb}}$. Based on [34], the STC model is formulated as:

$$\chi_{\mathrm{STC}}^{\mathrm{d}} \cdot A_{\mathrm{STC}}^{\mathrm{min}} \leq A_{\mathrm{STC}}^{\mathrm{d}} \leq \chi_{\mathrm{STC}}^{\mathrm{d}} \cdot A_{\mathrm{STC}}^{\mathrm{max}} \tag{2.7a}$$

$$\dot{\mathbf{Q}}_{\mathrm{STC,out}}^{\mathrm{d}}(i) = A_{\mathrm{STC}}^{\mathrm{d}} \cdot f_{\mathrm{STC}} \cdot \left(\mathbf{I}_{\mathrm{STC}}(i) - U_{\mathrm{STC}} \cdot \left(T_{\mathrm{STC}} - \mathbf{T}_{\mathrm{amb}}(i) \right) \right) \tag{2.7b}$$

$$A_{\mathrm{STC}}^{\max} = l_{\mathrm{BDG}} \cdot w_{\mathrm{BDG}}. \tag{2.7c}$$

In Eq. (2.7c), thermal losses of the collector are quantified by the term $U_{\mathrm{STC}} \cdot (T_{\mathrm{STC}} - \mathbf{T}_{\mathrm{amb}}(i))$, with U_{STC} being the thermal transmittance to the surroundings and T_{STC} denoting the temperature of the water entering the STC. The heat removal factor f_{STC} represents the efficiency.

2.4.3 Vapor Compression System

The vapor compression system (VCS) provides cooling either directly to the building or via an ice storage system (ISS). We aim to find a control-oriented model of the VCS which relates the required power input $\dot{\mathbf{E}}_{\mathrm{VCS,in}}$ needed to produce the required cooling power at a coolant temperature T_{c} and an ambient temperature T_{amb}. Fortunately, it is sufficient to develop a model only for T_{c}=−5°C and T_{c}=4°C, since the VCS supplies cooling power at these two temperatures for charging the ice storage tank or for directly cooling the building, respectively [39]; noting that these numbers depend on the building. This fact simplifies the modeling task considerably. Furthermore, the VCS is unable to provide cooling power with different T_{c} at the same time.

The data provided by the manufacturer together with the values of T_{c} and T_{amb} are used to determine the full-load coefficient of performance $\mathrm{COP}_{\mathrm{FL}}$. Figure 2.4 depicts the full-load COP as a function of the T_{amb} for the two constant coolant temperatures. Clearly, the figure shows a nonlinear behavior. Therefore, a linearized version is used in our implementations.

The cooling capacity of the VCS is denoted as $\dot{Q}'_{\mathrm{VCS}}$ which represents the maximum cooling power which can be supplied by VCS and is a design variable. Usually the value for $\dot{Q}'_{\mathrm{VCS}}$ is defined for the operating point T_{amb}=35°C and T_{c}=7°C. If the operating conditions are changed, the maximum available cooling power of the VCS changes accordingly. This new cooling power is called the cooling capacity at nonstandard conditions, denoted by $\dot{Q}'^{\mathrm{NS}}_{\mathrm{VCS}}$. Equation set (2.8) shows the calculation

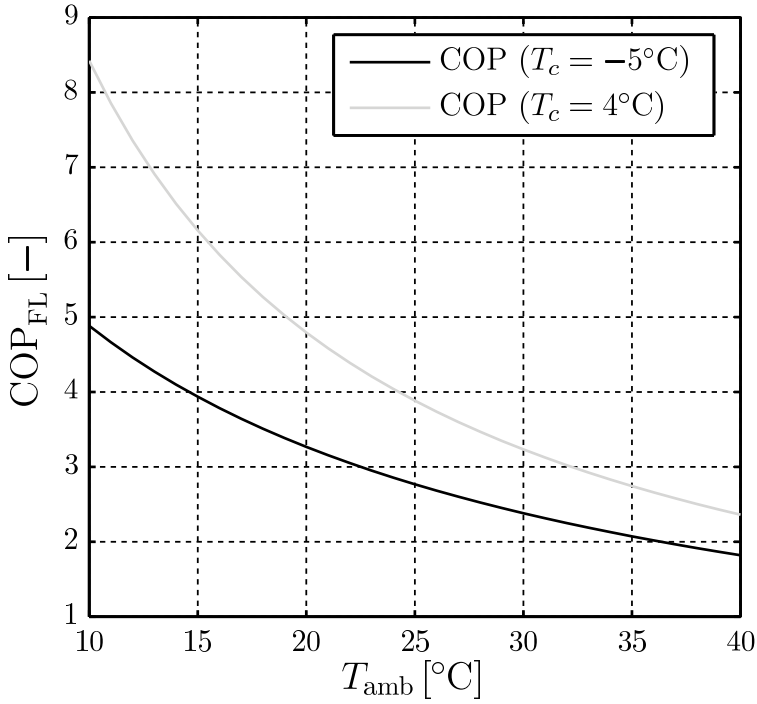


Figure 2.4: COP of VCS at full-load operation for two coolant temperatures.

of the total cooling power delivery to the building and to the ISS unit. Factors are used to compensate for the nonstandard conditions.

$$\dot{Q}'_{\mathrm{VCS}} = f^{\mathrm{NS}}_{\mathrm{BDG}} \cdot \dot{Q}'^{\mathrm{NS}}_{\mathrm{VCS}\to\mathrm{BDG}} + f^{\mathrm{NS}}_{\mathrm{ISS}} \cdot \dot{Q}'^{\mathrm{NS}}_{\mathrm{VCS}\to\mathrm{ISS}} \tag{2.8a}$$

$$f^{\mathrm{NS}}_{\mathrm{BDG}} = (1 + \pi_{\mathrm{amb}} \cdot (T_{\mathrm{amb}} - 35°\mathrm{C}) + \pi_c \cdot (T^{\mathrm{BDG}}_{\mathrm{c}} - 7°\mathrm{C})) \tag{2.8b}$$

$$f^{\mathrm{NS}}_{\mathrm{ISS}} = (1 + \pi_{\mathrm{amb}} \cdot (T_{\mathrm{amb}} - 35°\mathrm{C}) + \pi_c \cdot (T^{\mathrm{ISS}}_{\mathrm{c}} - 7°\mathrm{C})) \tag{2.8c}$$

The linear regression parameters π_{amb} and π_c are found using a linearization of COP curves in Fig. 2.4.

The last issue to be considered is the part-load operation. A part-load ratio PLR is defined to relate the actual part-load COP of the device to the full-load COP. This nonlinear equation is shown in Eq. (2.9).

$$\mathrm{COP}_{\mathrm{PL}} = \mathrm{COP}_{\mathrm{FL}} \cdot \frac{\mathrm{PLR}}{\pi^1_{\mathrm{PLR}} + \pi^2_{\mathrm{PLR}} \cdot \mathrm{PLR} + \pi^3_{\mathrm{PLR}} \cdot \mathrm{PLR}^2} \tag{2.9a}$$

$$= \mathrm{COP}_{\mathrm{FL}} \cdot {f^{\mathrm{PL}}_{\mathrm{VCS}}}^{-1} \tag{2.9b}$$

The part-load factor $f^{\mathrm{PL}}_{\mathrm{VCS}}$ depends on the type of the VCS. More specifically, the parameters π^1_{PLR}, π^2_{PLR} and π^3_{PLR} depend on various factors of the refrigeration machine such as the type of compressor. The average values for hermetic, reciprocating, and centrifugal compressors are shown in Table 2.1 which are taken from [40].

Table 2.1: Part-load parameters of Eq. (2.9) for three types of compressors.

	Compressor type		
	Hermetic	Reciprocating	Centrifugal
π^1_{PLR}	0.160	0.023	0.049
π^2_{PLR}	0.316	1.429	0.545
π^3_{PLR}	0.519	-0.471	0.389

Figure 2.5 depicts the COP at part-load operation for a VCS equipped with a hermetic compressor and a $\mathrm{COP}_{\mathrm{FL}}$ of 3.04.

As shown in Fig. 2.5, the part-load COP reaches its maximum value at a PLR of about 0.55, instead of 1. This is a typical behavior for VCS units which are equipped with hermetic compressors.

Finally, the VCS is implemented in our design framework using the following MILP equations:

$$\chi^{\mathrm{d}}_{\mathrm{VCS}} \cdot \dot{Q}'^{\min}_{\mathrm{VCS}} \leq \dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}} \leq \chi^{\mathrm{d}}_{\mathrm{VCS}} \cdot \dot{Q}'^{\max}_{\mathrm{VCS}} \tag{2.10a}$$

$$\dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}\to\mathrm{BDG}} + \chi^{\mathrm{d}}_{\mathrm{ISS}} \cdot \dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}\to\mathrm{ISS}} \leq \dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}} \tag{2.10b}$$

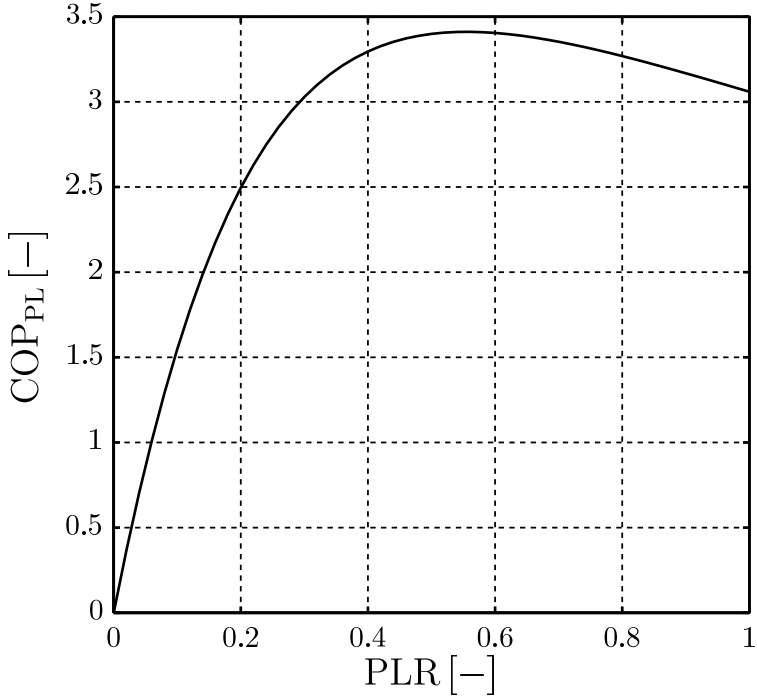


Figure 2.5: COP at part-load operation as a function of part-load ratio for a VCS with a hermetic compressor.

$$\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{VCS,in}} = \left(\frac{\chi^{\mathrm{d}}_{\mathrm{ISS}} \cdot \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{VCS}\to\mathrm{ISS}} \cdot f^{\mathrm{NS}}_{\mathrm{ISS}}}{\mathrm{COP}_{\mathrm{FL,VCS}\to\mathrm{ISS}}} + \frac{\dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{VCS}\to\mathrm{BDG}} \cdot f^{\mathrm{NS}}_{\mathrm{BDG}}}{\mathrm{COP}_{\mathrm{FL,VCS}\to\mathrm{BDG}}} \right) \cdot f^{\mathrm{PL}}_{\mathrm{VCS}} \quad (2.10c)$$

$$\dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}\to\mathrm{BDG}} = f^{\mathrm{NS}}_{\mathrm{BDG}} \cdot \dot{Q}'^{\mathrm{d,NS}}_{\mathrm{VCS}\to\mathrm{BDG}} \quad (2.10d)$$

$$\dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}\to\mathrm{ISS}} = f^{\mathrm{NS}}_{\mathrm{ISS}} \cdot \dot{Q}'^{\mathrm{d,NS}}_{\mathrm{VCS}\to\mathrm{ISS}} \quad (2.10e)$$

where the cooling power capacity, $\dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}}$ is the main design variable. Equation (2.10b) implies that the energy output during each time step is limited by the capacity of the VCS, while the cooling power is freely shared between the ISS and the building. The variable $\chi^{\mathrm{d}}_{\mathrm{ISS}}$ defines whether or not the ISS exists in the system. Equation (2.10c) describes the inlet electricity power as a function of the cooling power provided to

the building or the ISS, the two respected COP values. More details can be found in [39].

2.4.4 Heat Pumps

Heat Pumps basically have the same internal functionality as the VCS, except that they go through a reverse cycle: they remove the heat from the ambient and deliver it to the building envelope. Therefore, the MILP implementation of the heat pumps is very similar to the one of the VCS.

The main different is the definition of the COP. Obviously, the electric loss in a heat pump is an assistance to the heat output, implying a higher efficiency compared to the VCS. We use the concept of energy efficiency ratio (EER) as the equivalent COP if the heat pump was to be used for cooling. Hence, for the air-source heat pump (AHP) we have

$$\mathrm{EER}_{\mathrm{AHP}} = \frac{\dot{Q}'_{\mathrm{AHP}}}{\dot{E}_{\mathrm{AHP,in}}} \tag{2.11}$$

and

$$\mathrm{COP}_{\mathrm{AHP}} = \frac{\dot{Q}_{\mathrm{AHP}}}{\dot{E}_{\mathrm{AHP,in}}} \tag{2.12}$$

and thus resulting in

$$\mathrm{EER}_{\mathrm{AHP}} = \mathrm{COP}_{\mathrm{AHP}} - 1. \tag{2.13}$$

The Eq. (2.13) implies that the COP provided by the heat pump manufacturer must be reduced by 1 in order to find the EER for an equivalent VCS. Then, all the calculations explained in section 2.4.3 can be used for the heat pump. Furthermore, the heat storage will be, for example, a thermal energy storage (TES) such as a hot water tank.

Both an AHP and a ground source heat pump (GHP) are implemented in the framework. The AHP model is presented as:

$$\chi^{\mathrm{d}}_{\mathrm{AHP}} \cdot \dot{Q}^{\mathrm{min}}_{\mathrm{AHP}} \leq \dot{Q}^{\mathrm{d}}_{\mathrm{AHP}} \leq \chi^{\mathrm{d}}_{\mathrm{AHP}} \cdot \dot{Q}^{\mathrm{max}}_{\mathrm{AHP}} \tag{2.14a}$$

$$\dot{Q}^{\mathrm{d}}_{\mathrm{AHP}\rightarrow\mathrm{BDG}} + \chi^{\mathrm{d}}_{\mathrm{TES}} \cdot \dot{Q}^{\mathrm{d}}_{\mathrm{AHP}\rightarrow\mathrm{TES}} \leq \dot{Q}^{\mathrm{d}}_{\mathrm{AHP}} \tag{2.14b}$$

$$\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{AHP,in}} = \left(\frac{\chi^{\mathrm{d}}_{\mathrm{TES}} \cdot \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{AHP}\to\mathrm{TES}} \cdot f^{\mathrm{NS}}_{\mathrm{TES}}}{\mathrm{COP}_{\mathrm{FL,AHP}\to\mathrm{TES}}} + \frac{\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{AHP}\to\mathrm{BDG}} \cdot f^{\mathrm{NS}}_{\mathrm{BDG}}}{\mathrm{COP}_{\mathrm{FL,AHP}\to\mathrm{BDG}}} \right) \cdot f^{\mathrm{PL}}_{\mathrm{AHP}} \quad (2.14c)$$

$$\dot{Q}^{\mathrm{d}}_{\mathrm{AHP}\to\mathrm{BDG}} = f^{\mathrm{NS}}_{\mathrm{BDG}} \cdot \dot{Q}^{\mathrm{d,NS}}_{\mathrm{AHP}\to\mathrm{BDG}} \quad (2.14d)$$

$$\dot{Q}^{\mathrm{d}}_{\mathrm{AHP}\to\mathrm{TES}} = f^{\mathrm{NS}}_{\mathrm{TES}} \cdot \dot{Q}^{\mathrm{d,NS}}_{\mathrm{AHP}\to\mathrm{TES}} \quad (2.14e)$$

For a GHP, the temperature of the anergy source (ground) could be assumed constant. Depending on the geological situation, it usually varies between 0 and 10 °C.

2.4.5 Gas Boiler

The gas boiler (BGA) can be used in the areas with relatively cheap gas or oil prices and with milder restrictions on the CO_2 emissions. The main design variable is the outlet heating power capacity $\dot{Q}^{\mathrm{d}}_{\mathrm{BGA}}$.

$$\chi^{\mathrm{d}}_{\mathrm{BGA}} \cdot \dot{Q}^{\mathrm{min}}_{\mathrm{BGA}} \leq \dot{Q}^{\mathrm{d}}_{\mathrm{BGA}} \leq \chi^{\mathrm{d}}_{\mathrm{BGA}} \cdot \dot{Q}^{\mathrm{max}}_{\mathrm{BGA}} \quad (2.15a)$$

$$0 \leq \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BGA,out}}(i) \leq \dot{Q}^{\mathrm{d}}_{\mathrm{BGA}} \quad (2.15b)$$

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BGA,out}} = \eta_{\mathrm{BGA}} \cdot \dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{BGA,in}}. \quad (2.15c)$$

It is important to notice the difference between the scalar design variable $\dot{Q}^{\mathrm{d}}_{\mathrm{BGA}}$ which represents the optimal dimension of BGA (i.e. the maximum heating power) and the vector of optimal power output $\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BGA,out}}$, which has one value per sample time. Naturally, the latter is limited by the former, as shown in Eq. (2.15b). The η_{BGA} represents the efficiency of conversion from the vector of the chemical energy of fuel $\dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{BGA,in}}$ to the output heat.

2.4.6 Combined Heat and Power System

The combined heat and power systems (CHP) use various technologies such as engine-generator units and fuel cells. Generally, they consume one type of fuel and produce both heat and electricity to reach a higher overall efficiency [41]. In SBD, the inlet chemical power $\dot{H}^{\mathrm{d}}_{\mathrm{CHP}}$ is the

design variable of the CHP, since there are two output variables.

$$\chi^{\mathrm{d}}_{\mathrm{CHP}} \cdot \dot{H}^{\mathrm{min}}_{\mathrm{CHP}} \leq \dot{H}^{\mathrm{d}}_{\mathrm{CHP}} \leq \chi^{\mathrm{d}}_{\mathrm{CHP}} \cdot \dot{H}^{\mathrm{max}}_{\mathrm{CHP}} \tag{2.16a}$$

$$0 \leq \dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{CHP,in}}(i) \leq \dot{H}^{\mathrm{d}}_{\mathrm{CHP}} \tag{2.16b}$$

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{CHP,out}} = \eta_{\mathrm{CHP}} \cdot \dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{CHP,in}} \tag{2.16c}$$

$$\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{CHP,out}} = \eta_{\mathrm{CHP,el}} \cdot \dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{CHP,in}}. \tag{2.16d}$$

Note that these equations are only valid for full-load operation.

2.4.7 Absorption Chiller System

An absorption chiller system (ACS) is a promising cooling solution when a source of heat is also available, either because it is free (e.g., heat from an STC) or because it is a secondary product (e.g., heat from a CHP). The cooling power capacity $Q'^{\mathrm{d}}_{\mathrm{ACS}}$ is the main design variable.

$$\chi^{\mathrm{d}}_{\mathrm{ACS}} \cdot \dot{Q}'^{\mathrm{min}}_{\mathrm{ACS}} \leq \dot{Q}'^{\mathrm{d}}_{\mathrm{ACS}} \leq \chi^{\mathrm{d}}_{\mathrm{ACS}} \cdot \dot{Q}'^{\mathrm{max}}_{\mathrm{ACS}} \tag{2.17a}$$

$$0 \leq \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ACS,out}}(i) \leq \dot{Q}'^{\mathrm{d}}_{\mathrm{ACS}} \tag{2.17b}$$

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{ACS,in}} = \frac{\dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ACS,out}}}{\mathrm{COP}_{\mathrm{ACS}}} \tag{2.17c}$$

$$\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{ACS,in}} = f_{\mathrm{ACS,el}} \cdot \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ACS,out}}. \tag{2.17d}$$

Eq. (2.17c) calculates the input heating power based on the output cooling power and the COP. The required electricity input is also a factor of the output power, as shown in Eq. (2.17d).

2.5 Storage Block

The storage devices provide an energy buffer between the converters and the consuming devices. It is assumed that the three types of energy flow (heating, cooling, and electricity) are storable, while a storage of chemical energy is neglected. Figure 2.6 demonstrates the setup of storage devices and their interconnections to other parts of the system. As an

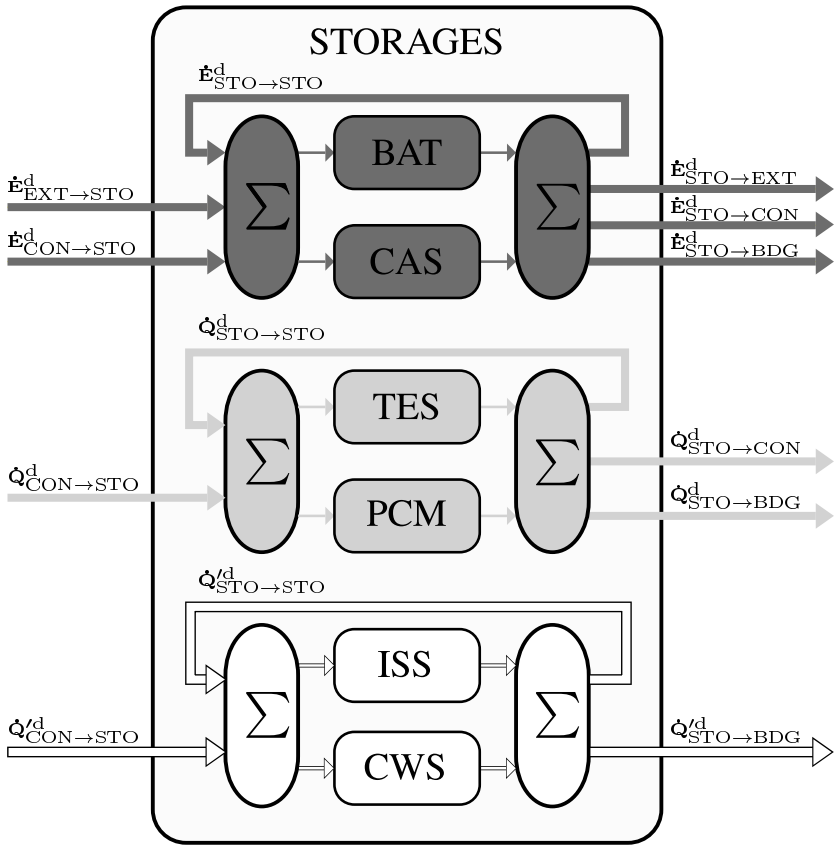


Figure 2.6: Overview of the external and internal connections of the storage block: electricity flow (dark grey), heating flow (light grey) and cooling flow (white).

example, the heat energy flows in the storage block are represented as:

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{CON}\to\mathrm{STO}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{STO}} = \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{TES,in}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{PCM,in}} \tag{2.18a}$$

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{EXT}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{STO}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{BDG}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{STO}\to\mathrm{CON}} = \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{TES,out}} + \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{PCM,out}}. \tag{2.18b}$$

Since different installed storage technologies may vary in capacity, available inlet/outlet power and self-discharge rate, it is possible in SBD to transfer the energy between storages (STO → STO). Although not efficient, this action enables storing the energy for long periods as well as meeting immediate high-power loads.

The following subsections describe the storage devices that are used in the SBD framework.

2.5.1 Battery Storage

Due to significant development effort being made in the technology of battery storage (BAT), it is foreseeable that the use of batteries in a building soon will become a possible option. The BAT model used in SBD is a control-oriented model. The battery degradation is neglected for simplicity. Also, there exist battery types such as Redox flow-batteries which show no degradation due to charge/discharge cycles [42]. The main design variable is the battery capacity $C^{\mathrm{d}}_{\mathrm{BAT}}$, which limits the amount of energy $\mathbf{E}_{\mathrm{BAT}}$ stored at any given time; as shown in Eq. (2.19b).

$$\chi^{\mathrm{d}}_{\mathrm{BAT}} \cdot C^{\mathrm{min}}_{\mathrm{BAT}} \leq C^{\mathrm{d}}_{\mathrm{BAT}} \leq \chi^{\mathrm{d}}_{\mathrm{BAT}} \cdot C^{\mathrm{max}}_{\mathrm{BAT}} \tag{2.19a}$$

$$0 \leq \mathbf{E}_{\mathrm{BAT}}(i) \leq C^{\mathrm{d}}_{\mathrm{BAT}} \tag{2.19b}$$

$$0 \leq \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{BAT,in}}(i) \leq \gamma_{\mathrm{BAT,ch}} \cdot C^{\mathrm{d}}_{\mathrm{BAT}} \tag{2.19c}$$

$$0 \leq \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{BAT,out}}(i) \leq \gamma_{\mathrm{BAT,dch}} \cdot C^{\mathrm{d}}_{\mathrm{BAT}} \tag{2.19d}$$

$$\begin{aligned} \mathbf{E}_{\mathrm{BAT}}(i) = &\left(1 - \sigma_{\mathrm{BAT}}\right) \cdot \mathbf{E}_{\mathrm{BAT}}(i-1) \\ &+ t_{\mathrm{s}} \cdot \left(\eta_{\mathrm{BAT,ch}} \cdot \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{BAT,in}}(i-1) - \frac{1}{\eta_{\mathrm{BAT,dch}}} \cdot \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{BAT,out}}(i-1) \right) , \\ &, \; i = 2, 3, ..., N. \end{aligned} \tag{2.19e}$$

Using the coefficients $\gamma_{\mathrm{BAT,ch}}$ and $\gamma_{\mathrm{BAT,dch}}$, the maximum charging and discharging powers are adaptively defined based on the capacity, as shown in Eq. (2.19c) and (2.19d). The state-of-charge or stored energy vector is calculated using Eq. (2.19e), where σ_{BAT} is the self-discharge

rate and the constants $\eta_{\mathrm{BAT,ch}}$ and $\eta_{\mathrm{BAT,dch}}$ represent the battery charging and discharging efficiencies, respectively.

2.5.2 Thermal Energy Storage

The thermal energy storage (TES), e.g., a hot water tank, is modeled analogously to the BAT. The design variable is the energy storage capacity $C^{\mathrm{d}}_{\mathrm{TES}}$.

$$\chi^{\mathrm{d}}_{\mathrm{TES}} \cdot C^{\mathrm{min}}_{\mathrm{TES}} \leq C^{\mathrm{d}}_{\mathrm{TES}} \leq \chi^{\mathrm{d}}_{\mathrm{TES}} \cdot C^{\mathrm{max}}_{\mathrm{TES}} \tag{2.20a}$$

$$0 \leq \mathbf{Q}_{\mathrm{TES}}(i) \leq C^{\mathrm{d}}_{\mathrm{TES}} \tag{2.20b}$$

$$0 \leq \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{TES,in}}(i) \leq \gamma_{\mathrm{TES,ch}} \cdot C^{\mathrm{d}}_{\mathrm{TES}} \tag{2.20c}$$

$$0 \leq \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{TES,out}}(i) \leq \gamma_{\mathrm{TES,dch}} \cdot C^{\mathrm{d}}_{\mathrm{TES}} \tag{2.20d}$$

$$\begin{aligned} \mathbf{Q}_{\mathrm{TES}}(i) = &\left(1 - \sigma_{\mathrm{TES}}\right) \cdot \mathbf{Q}_{\mathrm{TES}}(i-1) \\ &+ t_{\mathrm{s}} \cdot \left(\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{TES,in}}(i-1) - \dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{TES,out}}(i-1) \right) , \\ &, \; i = 2, 3, ..., N. \end{aligned} \tag{2.20e}$$

The main difference is that the transmission losses are considered in the building model (η_{BDG}), as mentioned above.

2.5.3 Ice Storage System

The ice storage system (ISS) is modeled as a receiver consisting of water and ice in thermal equilibrium. The ISS is modeled similarly to the TES. Its main design variable is the energy storage capacity $C^{\mathrm{d}}_{\mathrm{ISS}}$.

$$\chi^{\mathrm{d}}_{\mathrm{ISS}} \cdot C^{\mathrm{min}}_{\mathrm{ISS}} \leq C^{\mathrm{d}}_{\mathrm{ISS}} \leq \chi^{\mathrm{d}}_{\mathrm{ISS}} \cdot C^{\mathrm{max}}_{\mathrm{ISS}} \tag{2.21a}$$

$$0 \leq \mathbf{Q}'_{\mathrm{ISS}}(i) \leq C^{\mathrm{d}}_{\mathrm{ISS}} \tag{2.21b}$$

$$0 \leq \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ISS,in}}(i) \leq \gamma_{\mathrm{ISS,ch}} \cdot C^{\mathrm{d}}_{\mathrm{ISS}} \tag{2.21c}$$

$$0 \leq \dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ISS,out}}(i) \leq \gamma_{\mathrm{ISS,dch}} \cdot C^{\mathrm{d}}_{\mathrm{ISS}} \tag{2.21d}$$

$$\begin{aligned}\mathbf{Q}'_{\mathrm{ISS}}(i) &= \left(1-\sigma_{\mathrm{ISS}}\right)\cdot\mathbf{Q}'_{\mathrm{ISS}}(i-1)\\ &+ t_{\mathrm{s}}\cdot\left(\dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ISS,in}}(i-1)-\dot{\mathbf{Q}}'^{\mathrm{d}}_{\mathrm{ISS,out}}(i-1)\right),\\ &, \; i=2,3,...,N.\end{aligned} \tag{2.21e}$$

The use of an ISS is sensible if, for example, a down-sizing of the VCS is desired [39].

2.5.4 Other Devices

The modular structure of SBD easily enables the user to introduce new devices and services into the building. We present a brief formulation guide for several additional components that appear in Figure 2.1. They could be suitable for certain applications, but are not included in the analyses.

2.5.5 Electric Boiler

The use of an electric heating element (BEL) is almost never suggested by the SBD, unless a more efficient device such as an AHP or a CHP cannot be employed. The BEL can be modeled similarly to a BGA, except that Eq. (2.15c) is replaced by:

$$\dot{\mathbf{Q}}^{\mathrm{d}}_{\mathrm{BEL,out}} = \eta_{\mathrm{BEL}}\cdot\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{BEL,in}}. \tag{2.22}$$

2.5.6 Integrated Wind Turbine

An integrated wind turbine (IWT) as a source of renewable energy may replace or assist the PVS. A control-oriented model of an IWT is derived as follows:

$$\chi^{\mathrm{d}}_{\mathrm{IWT}}\cdot A^{\mathrm{min}}_{\mathrm{IWT}} \leq A^{\mathrm{d}}_{\mathrm{IWT}} \leq \chi^{\mathrm{d}}_{\mathrm{IWT}}\cdot A^{\mathrm{max}}_{\mathrm{IWT}} \tag{2.23a}$$

$$\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{IWT,out}} = 0.5\cdot A^{\mathrm{d}}_{\mathrm{IWT}}\cdot\eta_{\mathrm{IWT,gen}}\cdot f_{\mathrm{IWT}}\cdot v^{3}_{\mathrm{wind}}\cdot\rho_{\mathrm{amb}} \tag{2.23b}$$

$$P_{\mathrm{IWT}} = a_{\mathrm{IWT}}\cdot A^{\mathrm{d}}_{\mathrm{IWT}} + b_{\mathrm{IWT}}\cdot\chi^{\mathrm{d}}_{\mathrm{IWT}} \tag{2.23c}$$

$$O_{\mathrm{IWT}} = P_{\mathrm{IWT}} \tag{2.23d}$$

where $\eta_{\mathrm{IWT,gen}}$ is the generator efficiency, f_{IWT} the power coefficient, ρ_{amb} the air density, $A^{\mathrm{d}}_{\mathrm{IWT}}$ the cross- sectional area of the turbine, and v_{wind} the wind speed. The power coefficient is a function of the wind speed and can be precalculated using wind speed and manufacturer data.

2.5.7 Cool Water Storage

A cool water storage (CWS) basically follows the same concepts as those of a TES. Therefore, Eq. (2.20) may be used for the CWS as well.

2.5.8 Compressed Air Storage

A model of a compressed air storage (CAS) is very similar to the BAT model shown in Eq. (2.19). The main design variable would be the CAS volume $V^{\mathrm{d}}_{\mathrm{CAS}}$, but in order to be consistent with the BAT model, a CAS capacity C_{CAS} can be defined and calculated using the ideal gas equation:

$$C_{\mathrm{CAS}} = V^{\mathrm{d}}_{\mathrm{CAS}} \cdot \mathrm{Pa}^{\mathrm{max}}_{\mathrm{CAS}} \cdot \ln\left(\frac{\mathrm{Pa}^{\mathrm{max}}_{\mathrm{CAS}}}{\mathrm{Pa}_{\mathrm{amb}}}\right) \tag{2.24}$$

for which the nonlinear part is precalculated and therefore used as a constant. Additionally, the efficiency of CAS is a function of the storage pressure level.

2.5.9 Phase Changing Material Thermal Storage

In case a phase changing material (PCM) is of interest for a certain building, a model similar to that of ISS can be used. Some comparisons of different PCM techniques available are provided in [43].

3 Simultaneous Deterministic Design and Control

3.1 Introduction

This chapter discusses the formulation of the deterministic design and control problem. Before the main optimization criterion is explained, a description of the electricity tariff schemes is provided. This is due to the significance of the tariff schemes in the concepts related to the smart grids, such as building-gird interaction and zero energy buildings [44].

3.2 Electricity Tariffs Schemes

Currently, the two-level electricity tariff is the most common tariff scheme, while in some countries real-time variable tariffs for the end-user are being introduced [45]. Both of these electricity tariffs are implemented in the SBD framework, since they enable the investigation of the effect of dynamic price fluctuations on the building design.

3.2.1 Two-level Electricity Tariff

In most countries, electricity tariff schemes use a two-level price model, where during the nighttime the low-level price, and during the daytime the high-level price is applied.

3.2.2 Variable Electricity Tariff

The variable tariff is generated using the model presented by Ulbig and Andersson in [46], although the SBD includes an extended version which

Table 3.1: Two-Level Electricity Tariffs

Tariff parameter	Time of day	Pricing
P_{el}	10pm -6am	0.05 €/kWh
	6am -10pm	0.11 €/kWh
$P_{\mathrm{el}}^{\mathrm{peak}}$	6am -10pm	0.08 €/kW

considers the effect of power-regulation prices in addition, formulated as:

$$P_{\mathrm{el}} = \left(\pi_{\mathrm{el}}^{1} \cdot \frac{P_{\mathrm{spot}}}{\overline{P}_{\mathrm{spot}}} + \pi_{\mathrm{el}}^{2} \cdot \frac{load}{\overline{load}} + \pi_{\mathrm{el}}^{3}\right) \cdot \overline{P}_{\mathrm{el}} + P_{\mathrm{reg}} \cdot f_{\mathrm{reg}} \ . \tag{3.1}$$

In Eq. (3.1) the electricity price P_{el} is modeled as a function of the spot market price P_{spot}, the regulation price P_{reg} caused by grid stabilization costs, the level of demand in the electric grid, *load*, and the average electricity price $\overline{P}_{\mathrm{el}}$. The parameter f_{reg} is used as a scaling factor to adjust the effect of regulation prices. If a comparison to the two-level electricity tariff is desired, the average price $\overline{P}_{\mathrm{el}}$ should be the same. As a result, the overall electricity cost remains in the same range, while the effect of extra time variability is investigated. An analysis of the design sensitivity to $\overline{P}_{\mathrm{el}}$ is shown in the first case-study.

3.3 Optimization Criteria

The principal optimization problem of the SBD framework is to find the optimal selection and dimensioning of building components in order to minimize the total costs. Since the optimal design and the corresponding operating problem are not independent, the SBD uses an optimal control approach. Hence the design process is separated from the control problem. This means that if the suggested design is used, no other control strategy yields better results (i.e. a lower cost function). Vice versa, if the building is operated using the suggested optimal control strategy, no other component design is advantageous. Such an approach is referred to as a simultaneous optimization of control and design [47].

3.3.1 Objective Function

The optimization is performed for a full year, while the objective is extended to 20 years, which represents a typical lifetime of the building services. The total objective function to be minimized consists of three terms associated with investment, operating, and discomfort objectives, as well as a term representing the subsidy discounts for all devices:

$$O^{\mathrm{tot}} = O^{\mathrm{inv}} + O^{\mathrm{opr}} + O^{\mathrm{dcm}} - S. \tag{3.2}$$

The investment objective O^{inv} is calculated as a summation of the price variable P_{XYZ}, which indicates the overall cost of purchasing, installation, maintenance and replacement of an arbitrary device XYZ, as shown in Eq. (3.3a). An operational objective O^{opr} accounts for the total amount of electricity and gas consumption.

$$O^{\mathrm{inv}} = \sum_{\mathrm{XYZ}} P_{\mathrm{XYZ}} \tag{3.3a}$$

$$\begin{aligned} O^{\mathrm{opr}} &= O^{\mathrm{opr}}_{\mathrm{el}} + O^{\mathrm{opr}}_{\mathrm{gas}} = \\ &\mathrm{NPV}^{\mathrm{opr}} \cdot \Bigg(\left(\dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{EXT}\rightarrow\mathrm{BDG}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{EXT}\rightarrow\mathrm{STO}} + \dot{\mathbf{E}}^{\mathrm{d}}_{\mathrm{EXT}\rightarrow\mathrm{CON}} \right) \cdot \mathbf{P}_{\mathrm{el}} \\ &+ \left(\dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{EXT}\rightarrow\mathrm{BDG}} + \dot{\mathbf{H}}^{\mathrm{d}}_{\mathrm{EXT}\rightarrow\mathrm{CON}} \right) \cdot \mathbf{P}_{\mathrm{gas}} \Bigg) \end{aligned} \tag{3.3b}$$

$$O^{\mathrm{dcm}} = f(T^{\mathrm{d}}_{\mathrm{BDG}}, T^{\mathrm{set}}_{\mathrm{BDG}}, n^{\mathrm{occ}}) \ . \tag{3.3c}$$

In Eq. (3.3b), the net present value (NPV) concept is applied in order to transform the future operating costs to an equivalent present value, calculated as [48]:

$$\mathrm{NPV}^{\mathrm{opr}} = \frac{(1+r)^{\tau_{\mathrm{BDG}}} - 1}{r \cdot (1+r)^{\tau_{\mathrm{BDG}}}} \tag{3.4}$$

where r is the interest rate (e.g. 3%) and τ_{BDG} presents the time horizon of interest for the building. A discomfort objective O^{dcm} (Eq. (3.3c)) is considered in order to maintain the building temperature within a comfortable range. The O^{dcm} translates a departure of the temperature beyond specified boundaries into monetary costs and is basically a

Table 3.2: The design value D of each device.

	PVS	STC	VCS	AHP	BGA	CHP	ACS	BAT	TES	ISS
D	$A^{\mathrm{d}}_{\mathrm{PVS}}$	$A^{\mathrm{d}}_{\mathrm{STC}}$	$\dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}}$	$\dot{Q}^{\mathrm{d}}_{\mathrm{AHP}}$	$\dot{Q}^{\mathrm{d}}_{\mathrm{BGA}}$	$\dot{H}^{\mathrm{d}}_{\mathrm{CHP}}$	$\dot{Q}'^{\mathrm{d}}_{\mathrm{ACS}}$	$C^{\mathrm{d}}_{\mathrm{BAT}}$	$C^{\mathrm{d}}_{\mathrm{TES}}$	$C^{\mathrm{d}}_{\mathrm{ISS}}$
Unit	m^2	m^2	kW	kW	kW	kW	kW	kWh	kWh	kWh

function of the set-point and the actual building temperature, as well as of the number of occupants in the building n_{BDG}. An example of the discomfort cost function is shown in the case study.

3.3.2 Device Pricing

The device price P_{XYZ} is affected by the existence, χ, and the main design variable, D, of the device. A summary of the design variable D for different devices is shown in Table 3.2.

Equation (3.5) shows the calculation of the price for including a device XYZ in the building:

$$P_{\mathrm{XYZ}} = a_{\mathrm{XYZ}} \cdot \chi^{\mathrm{d}}_{\mathrm{XYZ}} + b_{\mathrm{XYZ}} \cdot D^{\mathrm{d}}_{\mathrm{XYZ}} \ . \tag{3.5}$$

A parameter a_{XYZ} defines the price for the existence of the device, while a parameter b_{XYZ} represents the relative price according to the design variable. They are generated using linear combinations of several costs, shown as functions g and h:

$$a_{\mathrm{XYZ}} = g\Big(P^{\mathrm{cap}}_{\mathrm{XYZ}} + P^{\mathrm{mnt}}_{\mathrm{XYZ}} \cdot \mathrm{NPV}^{\mathrm{opr}} + P^{\mathrm{rpl}}_{\mathrm{XYZ}} \cdot \mathrm{NPV}^{\mathrm{rpl}}_{\mathrm{XYZ}}\Big) \tag{3.6a}$$

$$b_{\mathrm{XYZ}} = h\Big(P^{\mathrm{cap}}_{\mathrm{XYZ}} + P^{\mathrm{mnt}}_{\mathrm{XYZ}} \cdot \mathrm{NPV}^{\mathrm{opr}} + P^{\mathrm{rpl}}_{\mathrm{XYZ}} \cdot \mathrm{NPV}^{\mathrm{rpl}}_{\mathrm{XYZ}}\Big) \tag{3.6b}$$

where $P^{\mathrm{cap}}_{\mathrm{XYZ}}$ is the capital cost and $P^{\mathrm{mnt}}_{\mathrm{XYZ}}$ is the maintenance costs of the device, affected by the operation NPV of Eq. (3.4). However, the cost of device replacement $P^{\mathrm{rpl}}_{\mathrm{XYZ}}$ requires a separate treatment, since building services have various operating lifetimes and may require several replacements during the lifetime of the project. The net present value

for replacement, $\mathrm{NPV}^{\mathrm{rpl}}_{\mathrm{XYZ}}$ is defined as:

$$\mathrm{NPV}^{\mathrm{rpl}}_{\mathrm{XYZ}} = \sum_{i=1}^{n^{\mathrm{rpl}}} \frac{1}{(1+r)^{i \cdot \tau_{\mathrm{XYZ}}}} \tag{3.7}$$

using the device lifetime τ_{XYZ} and the required number of replacements n^{rpl}.

3.3.3 Device Subsidies

The governmental subsidies are defined as a summation of the investment subsidy $S^{\mathrm{inv}}_{\mathrm{XYZ}}$ in case a device is included and the subsidy $S^{\mathrm{prp}}_{\mathrm{XYZ}}$ that is proportional to the size of the device:

$$S = \sum_{\mathrm{XYZ}} \left(S^{\mathrm{inv}}_{\mathrm{XYZ}} \cdot \chi^{\mathrm{d}}_{\mathrm{XYZ}} + S^{\mathrm{prp}}_{\mathrm{XYZ}} \cdot D^{\mathrm{d}}_{\mathrm{XYZ}} \right) + E_{\mathrm{PVS}} \cdot S^{\mathrm{opr}}. \tag{3.8}$$

Furthermore, the operational subsidy, S^{opr}, encourages an increased production of renewable energies, in this case the electric energy output of the PVS.

3.3.4 Global Constraints

As the legislation on emissions and on the energy consumption in buildings is becoming more stringent, buildings have to be operated both cost effective and in compliance with certain constraints. In the SBD formulation, such legislative restrictions are considered by introducing additional constraints in the optimization problem formulation. Here, two such global constraints are formulated, namely a limitation of annual CO_2 emissions and of energy consumption per square meter of conditioned area of the building.

The consumption of both gas and electricity introduces CO_2 emissions at some point. Some generalized data are provided in [49] which are used to calculate equivalent CO_2 emission factors for electricity ($f_{\mathrm{el}\rightarrow\mathrm{CO_2}}$) and gas ($f_{\mathrm{gas}\rightarrow\mathrm{CO_2}}$) for Europe. A maximum CO_2 emission constraint is

formulated as:

$$\frac{E^{\mathrm{yr}} \cdot f_{\mathrm{el}\rightarrow \mathrm{CO}_2} + H^{\mathrm{yr}} \cdot f_{\mathrm{gas}\rightarrow \mathrm{CO}_2}}{A_{\mathrm{BDG}}} \leq L_{\mathrm{CO}_2} \tag{3.9}$$

where L_{CO_2} [ton/m²a] represents the maximum CO_2 emission legislation, E^{yr} and H^{yr} are the annual electricity and natural gas energy consumption, respectively, and A_{BDG} is the conditioned area of the building. By examining all energy flows between the superior systems and the external sources, it is found that:

$$\begin{aligned} E^{\mathrm{yr}} = {} & E^{\mathrm{yr}}_{\mathrm{EXT}\rightarrow \mathrm{BDG}} + E^{\mathrm{yr}}_{\mathrm{EXT}\rightarrow \mathrm{STO}} + E^{\mathrm{yr}}_{\mathrm{EXT}\rightarrow \mathrm{CON}} \\ & - E^{\mathrm{yr}}_{\mathrm{STO}\rightarrow \mathrm{EXT}} - E^{\mathrm{yr}}_{\mathrm{CON}\rightarrow \mathrm{EXT}} \end{aligned} \tag{3.10}$$

$$H^{\mathrm{yr}} = H^{\mathrm{yr}}_{\mathrm{EXT}\rightarrow \mathrm{CON}}. \tag{3.11}$$

As concepts such as zero-energy buildings are being introduced by governments, it is also of interest to consider a constraint on the annual energy consumption. This constraint is formulated by removing the CO_2 emission factors in Eq. (3.9), introducing an energy consumption legislation L_{energy} [kWh/m²a], presented as:

$$\frac{E^{\mathrm{yr}} + H^{\mathrm{yr}}}{A_{\mathrm{BDG}}} \leq L_{\mathrm{energy}}. \tag{3.12}$$

The constraint of Eq. (3.12) ensures that all calculated optimal control strategies lead to a maximum annual energy consumption smaller than or equal to L_{energy}, within the building envelope.

3.4 Case Study: Commercial Building

In this section, the SBD framework is used to find the optimal design and control commands for a commercial building located in the city of Zurich, with a maximum occupancy of 300 people. For this study, the vector of electric demand $\dot{\mathbf{E}}_{\mathrm{BDG,eqp}}$ is pre-generated using a typical benchmark model of consumption profile for commercial buildings, developed by The Swiss Society of Engineers and Architects (SIA) [37]. The potentials of SBD are investigated by means of analyzing the sensitivity of the design

with respect to boundary conditions and global constraints. The optimization covers a whole year, i.e. 8760 hours. During all investigations, the parameter values of Table 3.3 are used.

For the building studied, the discomfort objective of Eq. (3.3c) is implemented as:

$$O^{\mathrm{dcm}} \geq P^{\mathrm{dcm}} \cdot n^{\mathrm{occ}} \cdot \left(\pi^{\mathrm{dcm}} \cdot \left(T_{\mathrm{BDG}} - T_{\mathrm{BDG}}^{\mathrm{set}} \right) - \Delta T \right) \tag{3.13}$$

where the discomfort price P^{dcm} translates the deviation of temperature from the set-point value to equivalent cost. The weighting parameter π^{dcm} and the offset ΔT are adjusted to generate the desired profile. The number of present occupants, n^{occ}, is variable with time and is precalculated considering the working hours during the weekdays, weekends and holidays. Figure 3.1 shows the resulting feasible hull for O^{dcm}, in an instant of time when 100 occupants are present in the building. The parameters P^{dcm}, π^{dcm} and ΔT are tuned to keep the building temperature within the desired range of 22–26 degrees Celsius. The user may manipulate these parameters to achieve different temperatures. The selected parameter values are shown within the figure. The six elements of the vectors shown in Figure 3.1 correspond to the six boundary lines of the convex hull.

In the first investigation, the global constraints are disabled, and optimizations are performed for various values of average electricity price, $\overline{P}_{\mathrm{el}}$. The effects of including global constraints are shown in the second investigation. Because of its transparency and its liberalized market, the actual electricity pricing data used in the SBD is from the German electric grid. The actual value of $\overline{P}_{\mathrm{el}}$ for Germany in 2010 was equal to 0.1 €/kWh [50], while it is suggested that in the foreseeable future, electricity prices will increase, at least within the European Union region [51]. Figure 3.2 depicts the selected components and their sizes for 8 different optimizations, while other devices are not selected and therefore are not shown. The exact design values are available in Table 3.4. The first notable result is that installing neither the storage devices nor the renewable sources is optimal under the electricity price range examined. In all cases, the components selected include one cooling and

Table 3.3: Parameter values used in the case study.

Para.	Value	Para.	Value	Para.	Value	Para.	Value	Para.	Value
t_s	3600 s	$\eta_{PVS,ref}$	0.14	a_{VCS}	3397 €	a_{CHP}	1224 €	σ_{TES}	0
T_{BDG}^{min}	293 K	π_{PVS}	0.001 1/K	b_{VCS}	0.47 €/kW	b_{CHP}	0.18 €/kW	$\gamma_{TES,ch}$	$0.5/t_s$
T_{BDG}^{max}	301 K	U_{PVS}	29 W/m²K	$\pi_{AHP,1}$	13.39	$f_{ACS,el}$	0.009	$\gamma_{TES,dch}$	$0.3/t_s$
η_{BDG}	0.95	f_{PVS}	0.9	$\pi_{AHP,2}$	-0.047	COP_{ACS}	0.75	a_{TES}	1000 €
η'_{BDG}	1	$T_{PVS,ref}$	289 K	$\pi_{AHP,3}$	1.109	a_{ACS}	50000 €	b_{TES}	16e-6 €/kWh
U_{BDG}	3614 W/K	a_{PVS}	0	$\pi_{AHP,4}$	0.012	b_{ACS}	0.20 €/kW	σ_{ISS}	0
C_{BDG}	1.9e9 J/kgK	b_{PVS}	750 €/m²	a_{AHP}	-4768 €	$\eta_{BAT,ch}$	0.9	$\gamma_{ISS,ch}$	$0.12/t_s$
l_{BDG}	50 m	T_{STC}	313 K	b_{AHP}	2.18 €/kW	$\eta_{BAT,dch}$	0.9	$\gamma_{ISS,dch}$	$0.25/t_s$
w_{BDG}	18 m	U_{STC}	5.3 W/m²K	η_{BGA}	0.85	σ_{BAT}	0.001	a_{ISS}	26900 €
τ_{BDG}	20 years	f_{STC}	0.6	a_{BGA}	1224 €	$\gamma_{BAT,ch}$	$0.2/t_s$	b_{ISS}	73e-7 €/kWh
P_{gas}	0.07 €/kWh	a_{STC}	0	b_{BGA}	0.03 €/kW	$\gamma_{BAT,dch}$	$0.15/t_s$	π_{el}^1	0.4
r	0.028	b_{STC}	340 €/m²	η_{CHP}	0.56	a_{BAT}	0	π_{el}^2	0.54
f_{reg}	0.15	f_{VCS}^{PL}	1.1	$\eta_{CHP,el}$	0.27	b_{BAT}	1e-4 €/kWh	π_{el}^3	0.06

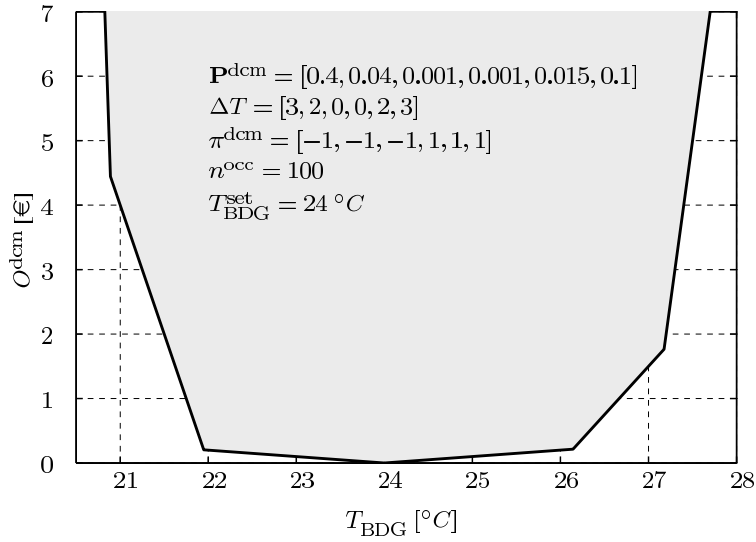


Figure 3.1: The feasible area (light grey) for the discomfort cost, as a function of building temperature.

two heating devices. The AHP is always chosen because it is an efficient device which takes advantage of anergy sources. For values of $\overline{P}_{\mathrm{el}}$ greater than 0.2 €/kWh, the gas is preferred as the main energy carrier. Therefore, the CHP is selected as the main provider of electricity, as the self-production is cheaper than the external supply. As a result, the ACS is chosen as the cooling device since there is enough waste heat available.

When the $\overline{P}_{\mathrm{el}}$ is smaller than 0.25 €/kWh, the selection of devices changes significantly. The CHP is replaced by a combination of BGA and a larger AHP, while the AHP is preferred as the price drops. Also in this price area, the cooling device is switched from ACS to VCS, which consumes only electricity. Obviously, manipulating the optimization parameters (such as device prices) result in a different switching point, but this sensitivity analysis qualitatively suggests that if the value of $\overline{P}_{\mathrm{el}}$ is increased and no other governmental restrictions are applied, it is reasonable to install CHP and ACS instead of BGA and VCS. Figure 3.3 demonstrates the heating power output of the CHP for two different de-

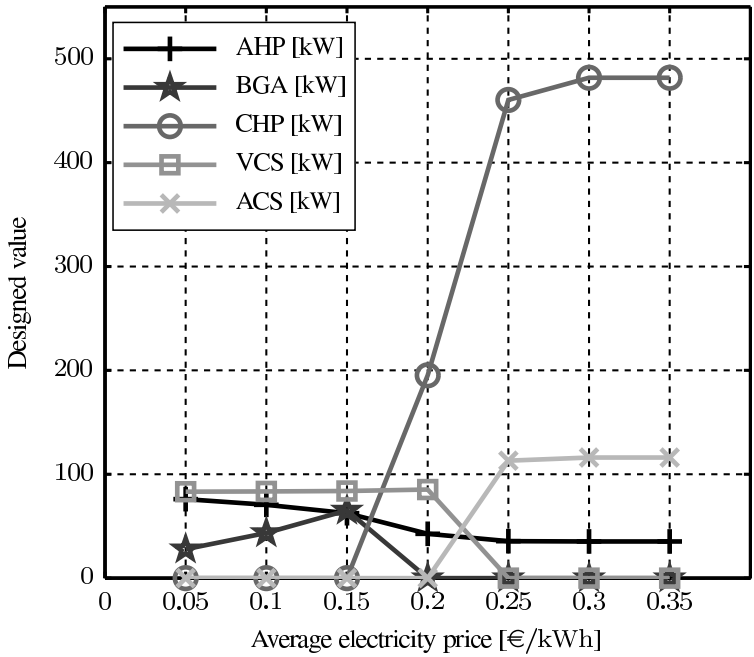


Figure 3.2: Component selection and dimensioning based on the average electricity price ($\overline{P}_{\mathrm{el}}$).

signs, corresponding to $\overline{P}_{\mathrm{el}} = 0.2$ and $\overline{P}_{\mathrm{el}} = 0.25$. It can be noticed that the CHP is very often, although not always, operated at its highest power capacity. One important issue with those episodic part-load operations is that most devices will work at highest efficiency only if operated within 50-100% of their nominal maximum power, implying that part-load operation is not desired. Although the final designs of SBD are based on such occasionally non-efficient behaviors, part-load operation can be avoided either by using energy storages, by replacing part-load operation with high-load duty cycles, or by splitting the demand to multiple devices. Furthermore, the start-up problem should be considered for actual operation, while it is beyond the scope of our investigations. In the second investigation, the value of $\overline{P}_{\mathrm{el}}$ is set to 0.1 €/kWh. Instead, different levels of the annual external energy consumption legislation, L_{energy}, are

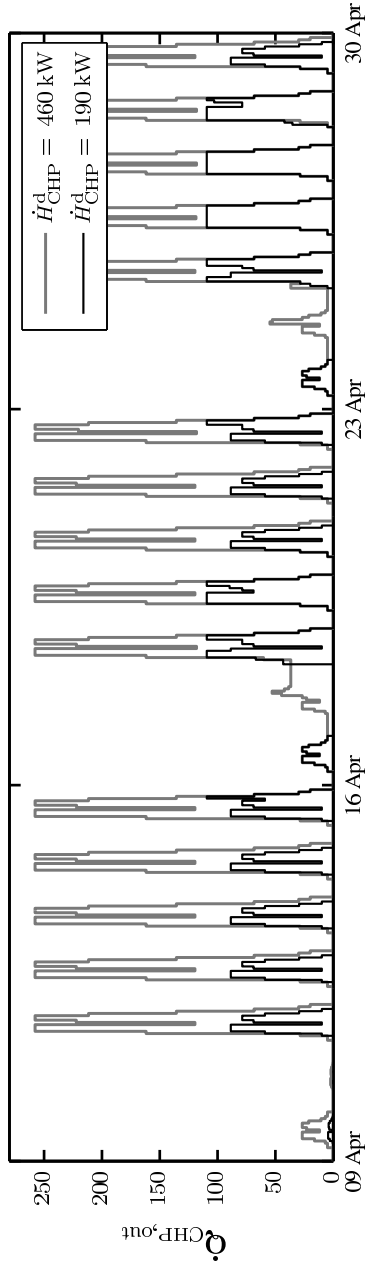


Figure 3.3: The heating power output of the CHP for three typical consequent weeks in April, shown for two different designs.

Table 3.4: Optimal design values for the first investigation.

	Design value [kW]				
$\overline{P}_{\text{el}}$	AHP	BGA	CHP	VCS	ACS
0.05	76.0	27.8	0	83.2	0
0.10	70.6	43.4	0	83.4	0
0.15	62.5	65.1	0	83.9	0
0.20	42.7	0	195	85.2	0
0.25	35.5	0	460	0	113
0.30	35.3	0	481	0	116
0.35	35.3	0	481	0	116

Table 3.5: Optimal design values for the second investigation.

	Design value [kW *or* kWh]							
L_{energy}	PVS	STC	AHP	BGA	VCS	ACS	TES	BAT
20	140	552	127	0	0	327	739	908
25	140	467	90.6	0	0	235	0	0
30	118	87.9	84.8	0	103	0	0	0
35	75.0	19.0	84.9	0	89.3	0	0	0
40	30.7	9.60	84.8	0	87.0	0	0	0
45	0	0	73.4	36.0	83.4	0	0	0
50	0	0	70.6	44.0	83.4	0	0	0
55	0	0	70.6	44.0	83.4	0	0	0

applied as the global constraints for the optimization. Figure 3.4 shows the resulting sizes of selected components, with exact values available in Table 3.5.

The first noticeable result is that, as before, the AHP is always chosen for the reasons mentioned above. For values of L_{energy} greater than 50 kWh/m^{2}a, the constraint is not affecting the optimization very much, hence the design is similar to the one shown in Figure 3.2. As the external consumption limit becomes tighter, the need for a more energy independent system emerges, since demand and supply must meet at all times. For a value of $L_{\text{energy}} < 40$, this need results in the introduction of PVS and STC devices. However, the installation of storage devices does

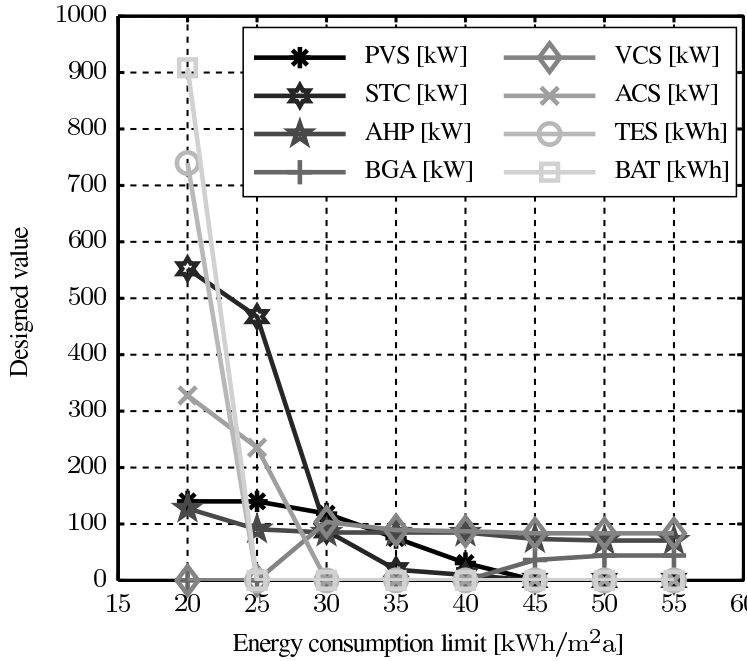


Figure 3.4: Component selection and dimensioning for different restrictions on maximum energy consumption.

not seem to be necessary until a very bounding constraint of $L_{\text{energy}} = 20$ is applied. Also, when the dimension of STC becomes large enough, the VCS is replaced by an ACS, due to the extra available heating power. All the analyses were performed on a computer with an Intel processor with a clock frequency of 2.66 GHz. The optimization for one year takes between 4 and 8 minutes.

As mentioned above, having an optimal design requires controlling the building in an optimal way. The SBD framework additionally enables the user to maintain the values of internal states and power flows within the desired boundaries, ensuring the comfort of occupants. Figure 3.5 shows the control for two typical days during the heating and cooling seasons, for the scenario of $L_{\text{energy}} = 25$ and $\overline{P}_{\text{el}} = 0.1$. The building uses the discomfort cost shown in Figure 3.1, where the building temperature

has a set-point of 24 °C. During the winter time, cooling devices are not used, while heating devices have to provide heat for the hot water demand throughout the year. During the day-light hours, there is usually a high amount of heating power available from the STC, while it is not always used by the building. This situation is allowed for in the SBD, namely the heating power dump, as shown in Eq. (2.3d). Also, since in this scenario there is no heat storage installed, the building mass is sometimes used to store the heat. This behavior is specially visible in the winter time, with the temperature increasing in the afternoon (Figure 3.5, top right). The phenomenon is avoidable by installing a TES.

3.4.1 Discussion

The potential for the optimization of building services is increased due to the development of renewable energy sources and storage technologies. In this thesis, a modular framework called the Smart Building Designer is described, which enables the optimal component design and control of a building, as well as sensitivity analyses over local or external parameters. The SBD can also be hired by lawmakers to investigate the effect of energy and environmental directives on the long-term behavior of residential and commercial buildings. Due to the flexible nature of the framework, arbitrary devices and energy carriers can be added by the user. Cumulative energy demand and life cycle carbon footprint can be assessed using the SBD framework. The investigations for the case studies show that the SBD is able to solve the optimization problem for one year in less than 10 minutes on a typical computer. This ability is mainly due to the accurate but control-oriented formulation of MILP models. The number of independent boolean variables are limited to the number of devices included, in order to speed up the optimization process even further. The SBD pre-calculates the required boundary conditions using the raw data such as those gained from meteorological measurements (temperature and solar irradiation), occupancy schedule, and spot-market electricity rates. As a result, the maximum flexibility is provided, enabling the user of SBD to apply the framework to an arbitrary building in any location. The sensitivity analyses also demon-

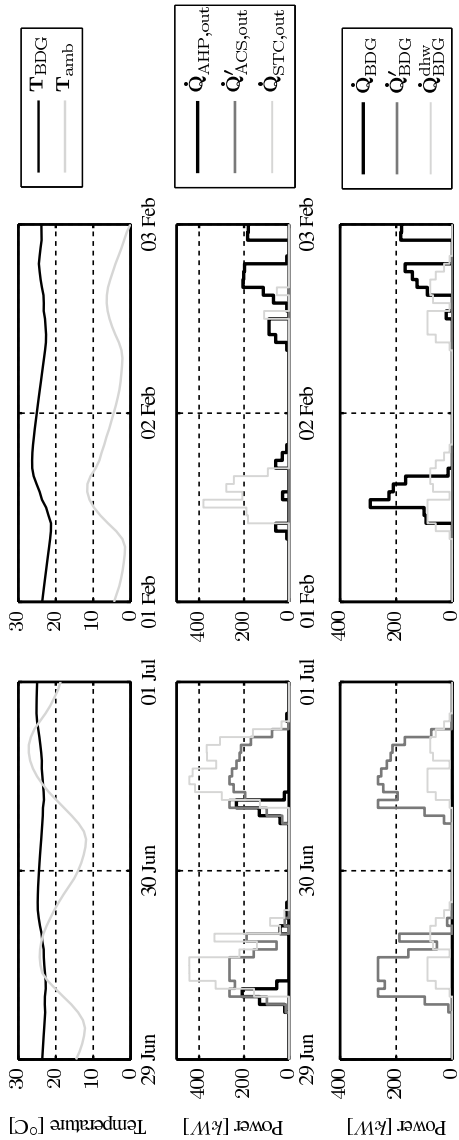


Figure 3.5: Building and ambient temperatures (top), available heating and cooling powers (middle), and actual demand for heating, cooling and domestic hot water (bottom) for two typical consequent days in winter (right) and summer (left).

strate how increasing energy-carrier prices or governmental legislation can result in different optimal designs.

3.5 Case Study: Cooling System with Storage

Buildings in general, and particularly their cooling systems, are responsible for high peak loads. Also, their associated costs of operation account for a considerable share of the total lifetime costs. As a consequence, their operating costs are highly sensitive to increasing energy prices. The integration of thermal energy storages, such as ice storage systems, is a promising solution to reduce the operating costs and the peak power consumption.

Several previous studies have already made important contributions to the optimal design and control of cooling systems with ice storage. In [52], an optimal design for a cooling system including cold storage facilities is presented, where the model-based operating strategy includes a pre-cooling of the building structure. A good introduction to the topic of thermal energy storage is provided in [53], where focus lies on the performance assessment of different ice storage systems in terms of energy efficiency. The authors in [54] present a model predictive control approach for a university cooling system and estimate the resulting electricity cost reductions at 24.5%. In [55], four different control strategies are compared using optimal control strategies as a benchmark, while in [56], the focus is on the optimal design of an ice storage system using particle swarm algorithms. Furthermore, the authors in [57] showed the optimal design for providing the grid with control reserve power.

In this case study, the cooling system includes a vapor compression system (VCS) and an ice storage system (ISS). Some simplifications are introduced to keep the models linear.

3.5.1 Parameters: Building

The building analyzed in this case study is a twelve-floor office building. Table 3.6 shows the building dimensions and relevant parameters.

The dynamic behavior of the building is modeled by a lumped parameter model of 1R1C as explained in section 2.3.3.

Table 3.6: Reference building data for the second case study.

Parameter	Value	Parameter	Value
w_{BDG}	16 m	C_{BDG}	1.75e9 J/K
l_{BDG}	50 m	$U_{\mathrm{BDG}}^{\mathrm{walls}}$	965.5 W/K
n_{occ}	907	$U_{\mathrm{BDG}}^{\mathrm{windows}}$	2.6e3 W/K
V_{BDG}	22 080 m^3		

3.5.2 Parameters: Vapor Compression System

The VCS is modeled as explained in the section 2.4.3. For this case study, data reported by [58] are used to determine the investment costs as a function of the capacity, as shown in Table 3.7.

Table 3.7: Investment costs of the VCS.

VCS Power Capacity	Investment Costs [€/kW]
$0\,\mathrm{kW} < \dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}} \leq 250\,\mathrm{kW}$	300
$250\,\mathrm{kW} < \dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}} \leq 500\,\mathrm{kW}$	280
$500\,\mathrm{kW} < \dot{Q}'^{\mathrm{d}}_{\mathrm{VCS}}$	240

3.5.3 Parameters: Ice Storage System

The ISS is also modeled as explained in the section 2.5.3. Table 3.8 shows the specific investment costs of ice storage system taken from several manufacturer data sheets for this case study.

Table 3.8: Investment costs of the ISS.

ISS Storage Capacity	Investment Costs
$C^{\mathrm{d}}_{\mathrm{ISS}} \leq 150$ kWh	5000 €
150 kWh $< C^{\mathrm{d}}_{\mathrm{ISS}} \leq$ kWh	30 €/kWh
2000 kWh $< C^{\mathrm{d}}_{\mathrm{ISS}}$	20 €/kWh

3.5.4 Results

For a given design, the control problem for an entire year is solved in less than one minute on a normal desktop computer. This low computational burden allows simulating the system for several promising system designs in the area of interest. The optimal designs for the two electricity tariff schemes investigated are shown here.

Two-level Tariff

In the two-level electricity tariff scheme, the optimization leads to a cooling system with a VCS capacity of 170 kW and an ISS storage capacity of 800 kWh for the given building. In contrast, the optimal system design without an ISS requires a VCS with a capacity of 230 kW. Figure 3.6 shows the total cost O^{tot} as a function of the ISS and the VCS capacity. Table 3.9 summarizes the optimization results and shows the various cost factors. Including an ISS in the cooling system, the investment costs of the VCS are reduced by 23 k€. Furthermore, the electricity costs are lowered by 17 k€. Together with the additional investment costs of the ISS, a cost reduction of 16 k€ is achieved. Additionally, the discomfort costs are lower for the cooling system with an ISS. Another interesting point is that the additional initial investment costs of the ISS are partly compensated by a reduced investment cost for the VCS.

The ISS helps to reduce demand charges since electricity power peaks are lowered. Figure 3.7 shows the operation of the optimal cooling sys-

Table 3.9: Results for the Two-Level Tariff

Cost Category	Cost with ISS	Cost without ISS
$O^{\mathrm{inv}}_{\mathrm{VCS}}$	67 k€	90 k€
$O^{\mathrm{inv}}_{\mathrm{ISS}}$	24 k€	0 k€
$O^{\mathrm{opr}}_{\mathrm{el}}$	73 k€	81 k€
$O^{\mathrm{peak}}_{\mathrm{el}}$	18 k€	27 k€
O^{dcm}	5 k€	11 k€
O^{tot}	187 k€	209 k€

tem design with ISS during the first three days of July. Obviously, the ISS is filled during nighttime when electricity prices and the ambient temperature are low. The stored cooling energy is then used to cover peak cooling demands in the afternoon of the next day. Furthermore, the cooling system pre-cools the building during nighttime to profit from the passive thermal energy storage of the building structure.

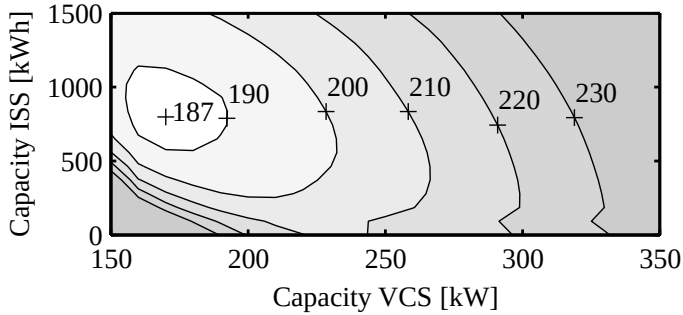


Figure 3.6: Total costs O^{tot} [k€] as a function of the VCS and the ISS capacities.

The cooling system covers the daily peak loads of about 440 kW by supplying 185 kW directly from the VCS to the building and by discharging the ISS by 255 kW. In contrast, the operation without ISS requires VCS loads of 255 kW during daytime. Analogously, the optimization routine makes use of pre-cooling the building structure in the early morning hours.

Variable Tariff

In the variable electricity tariff scheme, the optimal cooling system design consists of a VCS with a capacity of 190 kW and an ISS with a capacity of 790 kWh. The optimal dimension of the VCS without ISS is 270 kW. Table 3.10 provides the detailed results. The total cost reductions of 10 k€ achievable by using an ISS are rather moderate. Interestingly, the optimal capacity of the ISS is almost equal to the two-level tariff scheme. However, neglecting the demand charges and offering low or even negative working prices reduces the total lifetime costs by 32 k€ compared to the two-level tariff scheme.

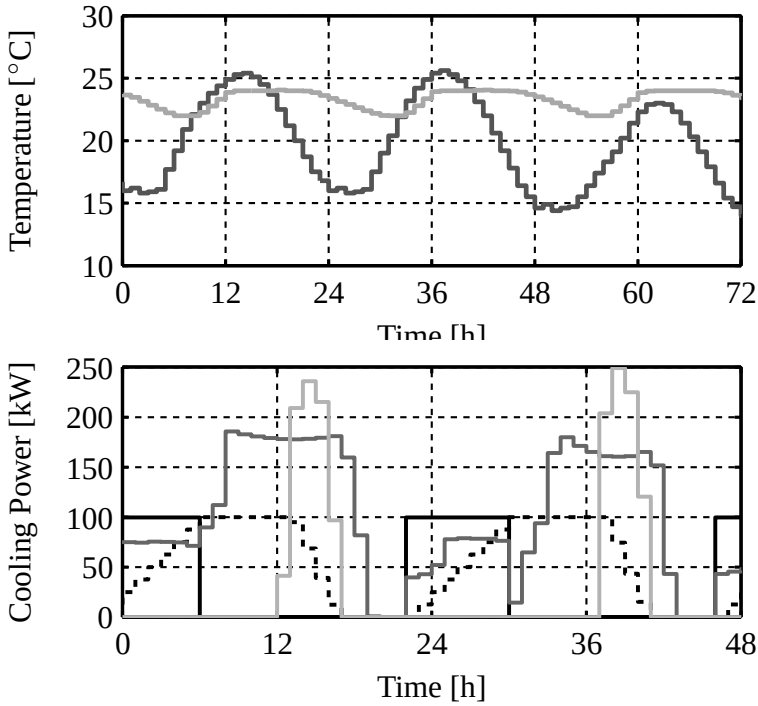


Figure 3.7: Cooling system operation using the two-level tariff during the first two days of July. Temperatures (top): building temperature (gray), ambient temperature (black). Cooling powers (bottom): ISS charging (black), direct building cooling (dark gray), building cooling by ISS (light gray), ice mass fraction (dotted).

The cooling system is indeed able to make active use of negative electricity prices as with ISS it consumes 9.9% and without ISS 13.5% of the electric power during these phases. Due to the larger VCS, the system without ISS has a better ability to profit from interactions with the power regulation market.

3.5.5 Discussion

Figure 3.8 depicts the optimal system operation during the first two days of July. Again, the ISS is mainly charged during nighttime when

Table 3.10: Optimization Cost Results ($f_{\mathrm{reg}} = 1$)

Cost Category	Cost with ISS	Cost without ISS
$O^{\mathrm{inv}}_{\mathrm{VCS}}$	75 k€	105 k€
$O^{\mathrm{inv}}_{\mathrm{ISS}}$	24 k€	0 k€
$O^{\mathrm{opr}}_{\mathrm{el}}$ when $P_{\mathrm{el}} \leq 0$	−1.8 k€	−2.4 k€
$O^{\mathrm{opr}}_{\mathrm{el}}$ when $P_{\mathrm{el}} > 0$	56 k€	60.0 k€
O^{dcm}	1.5 k€	2.3 k€
O^{tot}	155 k€	165 k€

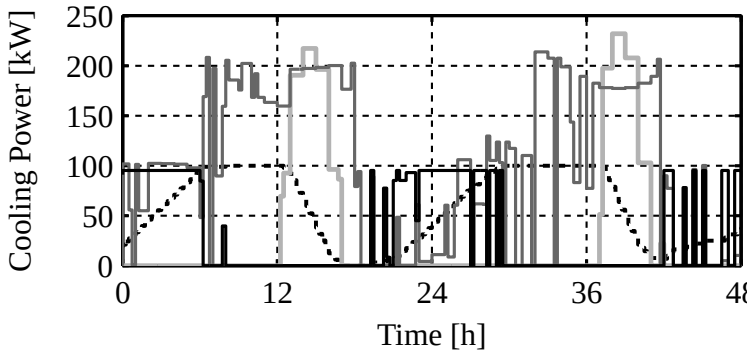


Figure 3.8: Cooling system operation during the first two days of July using a variable tariff with $f_{\mathrm{reg}} = 0.5$. Cooling powers: ISS charging (black), direct building cooling (dark gray), building cooling by ISS (light gray), ice mass fraction (dotted).

electricity prices are low and used to cover peak cooling demand in the daytime. Furthermore, the system interacts with the power regulation market by increasing or decreasing its cooling power depending on the current price. In contrast to the system operation under the two-level tariff regime, the operation load changes more frequently.

Figure 3.9 shows the total and the electricity cost as a function of the scaling parameter f_{reg} for the cooling system with and without ISS. Both costs decrease as the scaling factor increases. The cost saving potential of adding an ISS increases with rising price levels of the power regulation market. Interestingly, these results contradict the observation that a

cooling system without an ISS produces more cooling energy at low and negative prices. However, cooling systems with an ISS are able to avoid positive electricity price peaks more efficiently. As a consequence, cost-intensive operation during positive price peaks is less effectively avoided without an ISS.

Figure 3.10 shows the optimal VCS and the ISS design again as a function of the price scaling factor. Obviously, the optimal dimensioning of the cooling system is insensitive to the variability of the electricity price.

Clearly, it does not pay to increase the capacity of the VCS in order to benefit more from taking part in the control reserve market. The gains achieved by making use of low electricity prices are outweighed by increased investment and maintenance costs of the VCS.

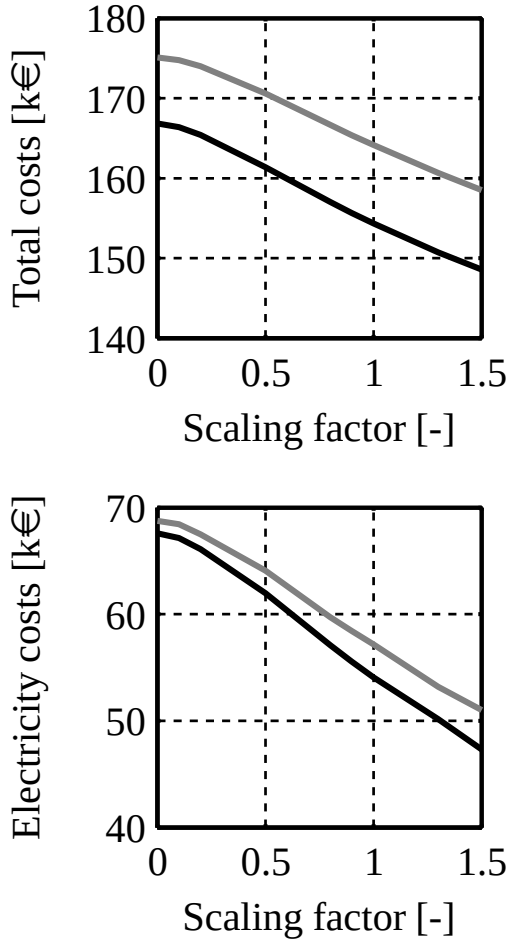


Figure 3.9: Total (top) and electricity cost (bottom) as a function of the regulation price scaling factor f_{reg}: with ISS (black), without ISS (gray).

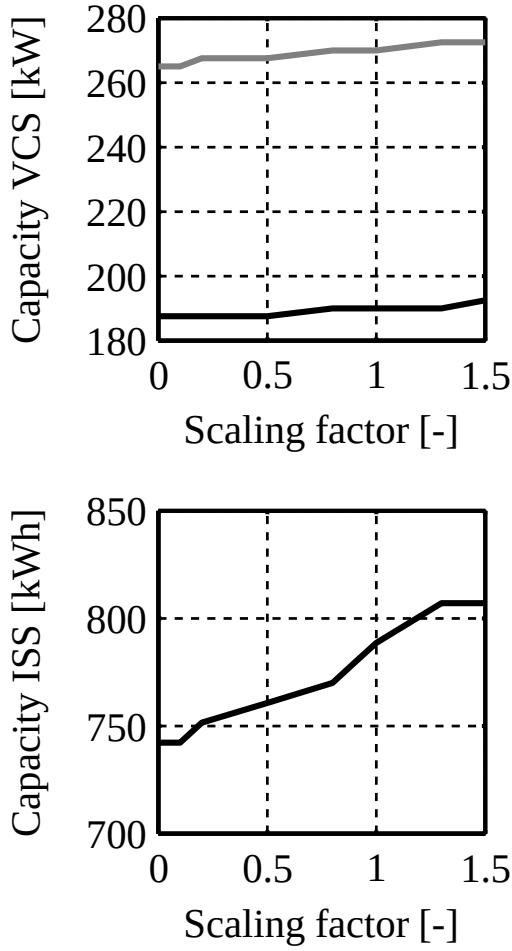


Figure 3.10: Optimal VCS capacity (top) and ISS capacity (bottom) as a function of the regulation price scaling factor f_{reg}: with ISS (black), without ISS (gray).

4 Uncertainty and Robust Design

4.1 Introduction

Several studies have targeted the problem of building system design, considering several subsystems such as the thermal [7], electricity generation and storage devices [9, 10] or a combination of these [12–14, 16, 17, 47] with the common approach being a deterministic optimization. The validity of the results in a deterministic model-based optimization is limited due to uncertainties in both the model parameters and the boundary conditions. The sensitivity of the system design with respect to these uncertainties may be significant, such that a slight change in the boundary condition might greatly influence the objective function. In order to overcome this problem, the uncertainties must be incorporated in the design process. This so-called robust design approach minimizes the risk of high operating and discomfort costs due to the unavoidable stochasticity. Several researchers have addressed the building system design problem in the presence of uncertainties [59–62].

Although the robust design proves to be the superior approach, the design sensitivity strongly depends on the system analyzed and on the variance of boundary conditions. For example, in [7] a robust design has been presented that uses the Monte Carlo method for uncertainties in the equipment gain, where a significant change in the design is noticed compared to the deterministic optimization results. In contrast, in a similar design problem described in [63], the outcomes of robust and deterministic approaches do not vary strongly. A sensitivity analysis over the energy pricing and carbon tax rates has been made in [64], proving a CO_2 emission reduction with increasing carbon tax and a design vari-

ation with increasing electricity pricing up to twice the actual cost. A very similar analysis was made in [63] by augmenting the consumption variance in the optimization; here the design becomes more robust, but only small differences are noticeable concerning the size of devices.

In this chapter, an extended version of the previously developed framework is presented to be able to account for uncertainties in the boundary conditions. Several relevant uncertain boundary conditions and their influence on the building system design are described. A number of sensitivity analyses are presented using a case study to investigate the influence of uncertainties related to the user behavior and external factors such as energy prices on the optimal design. Finally, the results of a robust design approach is compared with those obtained with the deterministic one to examine the effectiveness and computational efforts of each approach. Additionally, the set of deterministic sensitivity analyses is used as an assessment method to decide whether a robust design is beneficial.

4.2 Reduced Model Framework

The process of simultaneous design and selection of building services is based on a reduced version of the framework described in chapter 3. The framework is implemented in MATLAB using the modeling toolbox Yalmip [28] as an interface to the MILP solver. Figure 4.1 shows the modeled devices of the framework and the energy flows among them.

4.3 Stochastic Extension of the Objective Function

The objective described above only considers a deterministic case where all the boundary conditions are fixed. However, many boundary conditions are exposed to uncertainties, the most probable scenarios of which have to be considered in the design. Each scenario is characterized by a sample of possible combinations of values for uncertain boundary conditions. As a result, each scenario i corresponds to an objective function

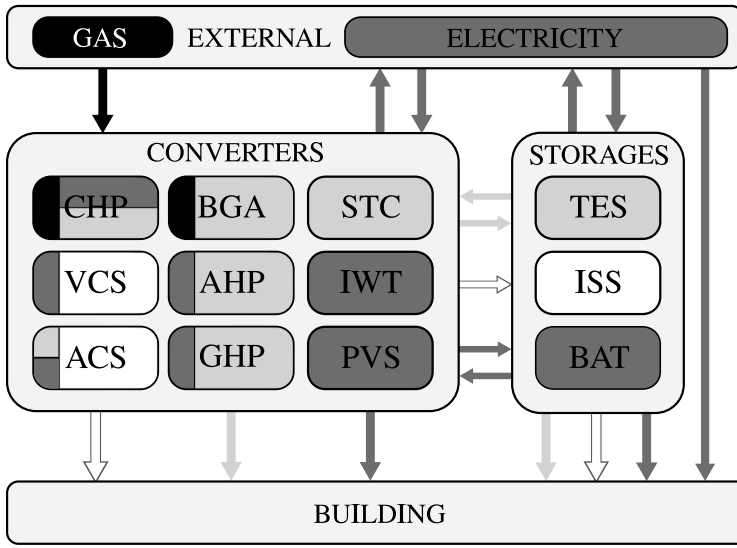


Figure 4.1: Overview of the reduced framework: gas flow (black), electricity flow (dark grey), heating flow (light grey) and cooling flow (white).

O_i^{tot} and a probability of occurrence p_i. However, only a cumulative objective can be optimized by the MILP solver.

The minimization of the overall expected cost is effected in a single step through convex optimization rather than with the commonly used Monte Carlo optimization. A straight-forward method is used to generate the stochastic objective function $O_{\text{stochastic}}^{\text{tot}}$, namely a minimization of the expected values of the objective functions of the individual scenarios:

$$O_{\text{stochastic}}^{\text{tot}} = \sum_{i=1}^{n_{\text{sc}}} p_i \cdot O_i^{\text{tot}} \quad \text{with} \quad \sum_{i=1}^{n_{\text{sc}}} p_i = 1 \tag{4.1}$$

where n_{sc} denotes the number of scenarios considered. Each scenario has an individual optimal operation strategy. A deterministic optimization is a special case of this formulation for $i = 1$. The objective function can also be extended as

$$O_i^{\text{tot}} = O^{\text{inv}} + O_i^{\text{opr}} + O_i^{\text{dcm}} + O_i^{\text{peak}} - S \tag{4.2}$$

where the O_i^{peak} represents the peak power cost. The resulting robust design consists of a single design of the building energy system and an individual control strategy for each scenario.

4.4 Uncertainties Considered

In order to simulate a set of possible relevant scenarios, several uncertain boundary conditions are considered to resemble a realistic stochasticity in the forecasts. Obviously, some parameters have a stronger effect on the final optimal design [65], which are taken into account for the present analysis.

4.4.1 Occupants and Equipment Gain

An increase in the number of people present in the building introduces two effects: first, an increased air-exchange rate, and second, higher internal gains. The number of people typically present in the building varies depending on the type of the building. For an office building the specific number is approximately 0.1 person/m^2 [37].

The equipment gain is defined as the heat produced by all electrical appliances in the building such as lights, computers and other electricity consuming devices. It varies from one building type to another. For an office building it is approximately 19 W/m^2 during the working hours [37].

4.4.2 Weather Conditions

The weather condition plays an important role in the design and operation of building devices, as its profile changes from one year to another. The most commonly utilized weather data is represented in the so-called typical meteorological years, a version of which is generated with Meteonorm software [66] and is used in this case study. In addition to a normal weather profile, an extreme one is generated containing more intense values of temperature, solar irradiation and wind speed, during both winter and summer. Figure 4.2 shows the monthly mean values of the ambient temperature and of average solar irradiation.

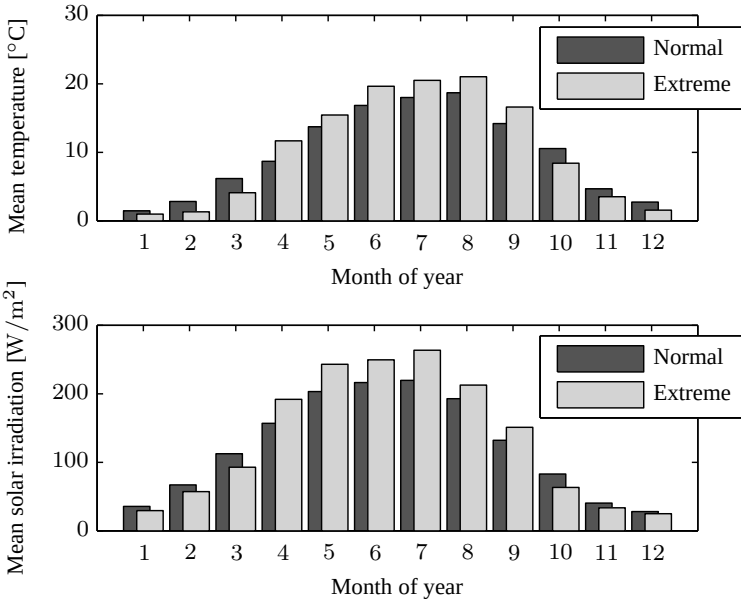


Figure 4.2: Extreme and normal weather profiles; Temperature profile (top) and solar irradiation profile (bottom).

4.4.3 Energy Price and Interest Rate

The energy pricing for gas and electricity is strongly influenced by geopolitical factors and may be subject to large variations. The high price variations, especially for fossil fuels, do not suggest a stability in energy price, leaving it a strongly uncertain variable in the building system design. In this thesis, a two-level electricity tariff and a fixed gas price are considered.

In all the analyses, the ratio between the the electricity and gas prices remains the same. The interest rate is mainly influenced by the national banks in order to tighten or ease the economic expansion with the goal to control inflation and to promote economic growth. The adjustment of the interest rate takes place periodically and is influenced by the economic situation of the respective countries [67]. In the design process, low interest rates boost investments in expensive and energy-efficient devices

which consume less electricity of fuel in average, while high interest rates result in lower investments but higher operating costs.

4.5 Case Study

4.5.1 Reference Deterministic Design

In the deterministic design case, for a given input of weather data, energy prices, building details, and devices available, a cost optimal design and its corresponding operation strategy throughout the year is generated. The output is computed using a convex optimization which minimizes the objective function consisting of costs for device installation, electricity, gas, occupants' discomfort and peak consumption.

In order to obtain a base scenario for comparing the deterministic and the robust approaches, a deterministic optimization is performed for a commercial building located in the city of Zurich. The optimization covers a whole year, i.e. 8760 hours.

Table 4.1: Results of a deterministic design. Devices which are not shown were not selected by the optimizer.

Device	PVS	AHP	CHP	VCS	TES
D	401	131	25	143	29
Unit	m^2	kW	kW	kW	kWh

The design output of the design framework is a single set of selected devices and their proper dimensions. All the devices shown in Fig. 4.1 may be chosen or not, hence the optimizer has to consider all the possible combinations of these devices. Table 4.1 shows the resulting design values obtained in the case of a deterministic approach, i.e. when no uncertainties exist on the boundary conditions. Table 4.2 shows the default settings and standard parameters.

It is observed that no storage device is selected, with the exception of a TES (thermal energy storage). Previous investigations have shown that the expensive BAT (battery) and ISS (ice storage system) should be included only if tight limitations on energy consumption are imposed.

Table 4.2: Default parameter values used in the case study.

Param.	Value	Param.	Value	Param.	Value	Param.	Value	Param.	Value
t_s	3600 s	$\eta_\mathrm{PVS,ref}$	0.14	a_VCS	3397 €	a_CHP	1224 €	σ_TES	0
$T^\mathrm{min}_\mathrm{BDG}$	293 K	π_PVS	0.001 $^1/_\mathrm{K}$	b_VCS	0.47 €/kW	b_CHP	0.18 €/kW	$\gamma_\mathrm{TES,ch}$	$0.5/t_\mathrm{s}$
$T^\mathrm{max}_\mathrm{BDG}$	301 K	U_PVS	29 $^\mathrm{W}/_{\mathrm{m^2K}}$	$\pi_\mathrm{AHP,1}$	13.39	f_IWT	1	$\gamma_\mathrm{TES,dch}$	$0.3/t_\mathrm{s}$
η_BDG	0.95	f_PVS	0.9	$\pi_\mathrm{AHP,2}$	-0.047	η_IWT	0.41	a_TES	1000 €
η'_BDG	1	$T_\mathrm{PVS,ref}$	289 K	$\pi_\mathrm{AHP,3}$	1.109	a_IWT	475 €	b_TES	16e-6 €/kWh
U_BDG	3614 $^\mathrm{W}/_\mathrm{K}$	a_PVS	0	$\pi_\mathrm{AHP,4}$	0.012	b_IWT	2.58 €/kW	σ_ISS	0
C_BDG	1.9e9 $^\mathrm{J}/_\mathrm{kgK}$	b_PVS	750 €/m^2	a_AHP	-4768 €	$\eta_\mathrm{BAT,ch}$	0.9	$\gamma_\mathrm{ISS,ch}$	$0.12/t_\mathrm{s}$
l_BDG	50 m	T_STC	313 K	b_AHP	2.18 €/kW	$\eta_\mathrm{BAT,dch}$	0.9	$\gamma_\mathrm{ISS,dch}$	$0.25/t_\mathrm{s}$
w_BDG	18 m	U_STC	5.3 $^\mathrm{W}/_{\mathrm{m^2K}}$	η_BGA	0.85	σ_BAT	0.001	a_ISS	26900 €
τ_BDG	20 years	f_STC	0.6	a_BGA	1224 €	$\gamma_\mathrm{BAT,ch}$	$0.2/t_\mathrm{s}$	b_ISS	73e-7 €/kWh
P_gas	0.07 €/kWh	a_STC	0	b_BGA	0.03 €/kW	$\gamma_\mathrm{BAT,dch}$	$0.15/t_\mathrm{s}$		
r	0.028	b_STC	340 €/m^2	η_CHP	0.56	a_BAT	0		
$\overline{P}_\mathrm{el}$	0.1 €/kWh	$f^\mathrm{PL}_\mathrm{VCS}$	1.1	$\eta_\mathrm{CHP,el}$	0.27	b_BAT	1e-4 €/kWh		

Due to a demand for domestic hot water, a TES (thermal energy storage) is always selected and is sized at its minimum allowed dimension. Therefore, the design value of a TES is not mentioned in the rest of this section. In order to investigate the change in the optimal design in the presence of uncertainties, a set of sensitivity analyses with respect to the most relevant boundary conditions are carried out.

4.5.2 Sensitivity Analysis

Each slight stochasticity in a boundary condition potentially affects the optimal design of components. However, the sensitivity of the design with respect to different parameters is not the same. In order to investigate the sensitivity, the following parameters are taken into account: The average number of occupants, the equipment gain, the weather condition, the annual interest rate and the energy prices.

Occupants and Equipment Gain

The impact of uncertainty in the number of occupants n^{occ} and of the equipment gain $\dot{\mathbf{E}}_{\mathrm{BDG,eqp}}$ on the design of the building devices is studied in this section. The values for the occupants' presence and the equipment gain are pre-generated using a typical benchmark model for commercial buildings, as developed by The Swiss Society of Engineers and Architects [37].

Figure 4.3 shows two sets of deterministic designs for reasonable ranges of the equipment gain and the number of occupants, while all other boundary conditions are the same for every design. Only the three devices AHP, VCS (vapor compression system) and CHP (combined heat and power system) were finally selected by the optimizer. The normalized dimensions are shown for the comparison. Two important observations can be made here. First, the dimension of each device changes smoothly in the uncertainty range studied. Second, the list of selected and non-selected devices remains the same. As a rule of thumb, one may choose the maximum suggested dimension for the selected devices in a practical range of uncertainties to guarantee the comfort of the occupants. Then, a change in the boundary conditions are compensated by a different operating strategy.

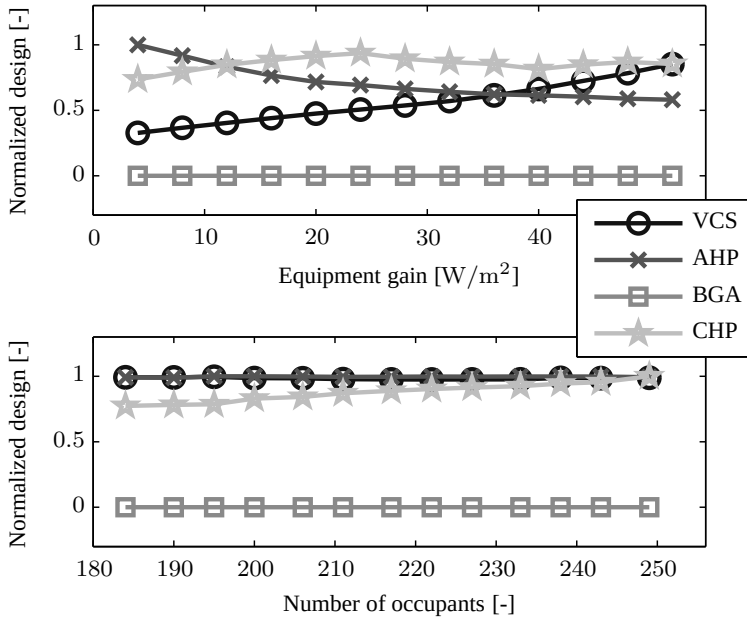


Figure 4.3: Deterministic designs for different values of the equipment gain (top) and the number of occupants (bottom).

Weather

The impact of different weather conditions on the design of building devices using the design tool is shown in Fig. 4.4. Heating and cooling devices are over-dimensioned for extreme weather conditions in order to handle the higher temperature deviations both in the summer and winter time, as shown in Fig. 4.2. The increased dimensions of AHP, VCS and CHP clearly show this behavior. This higher nominal powers results in an overall increase of the electricity consumption. Because of the peak power costs, for extreme weather conditions the BGA (gas boiler) is replaced by a CHP which additionally produces electricity, thus avoiding the high peak costs.

When both weather conditions are taken into account to form a stochastic objective function, the resulting optimal design is almost an average of the deterministic designs, which implies an almost linear behavior.

However, when compared to the previous boundary conditions, an obvious difference is the change of device selection, as no gas boiler is selected for the extreme weather conditions. In such scenarios, a robust design approach might be beneficial for finding a more robust optimal design.

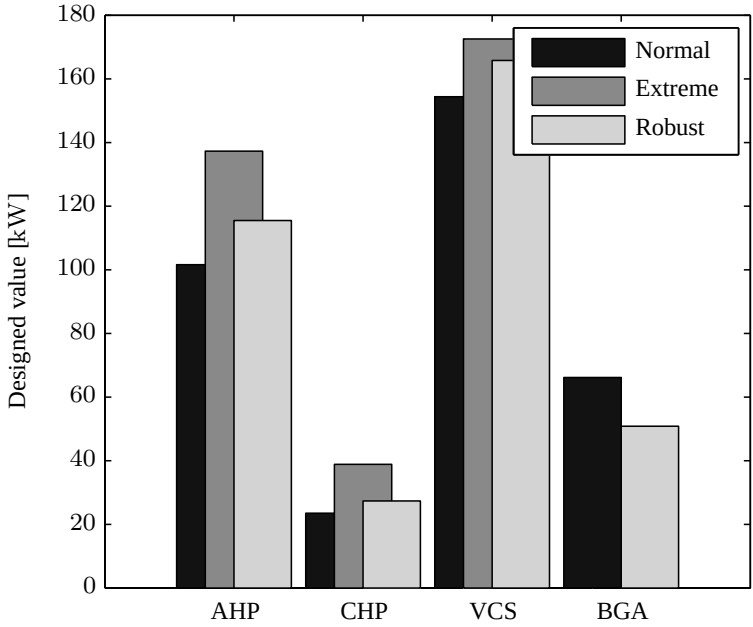


Figure 4.4: The changes in the optimal dimensions shown for two deterministic designs based on normal and extreme weather conditions. Furthermore, a robust design is performed to combine the two weather conditions.

Energy Price and Interest Rate

The price for energy carriers and annual interest rate both have a connection to the economy. While the uncertainty of these boundary conditions is not as strong as that of weather forecasts, they are still of interest because of the relevant impact on the long-term operating and investment costs. The energy price is calculated as a mean of the average electricity

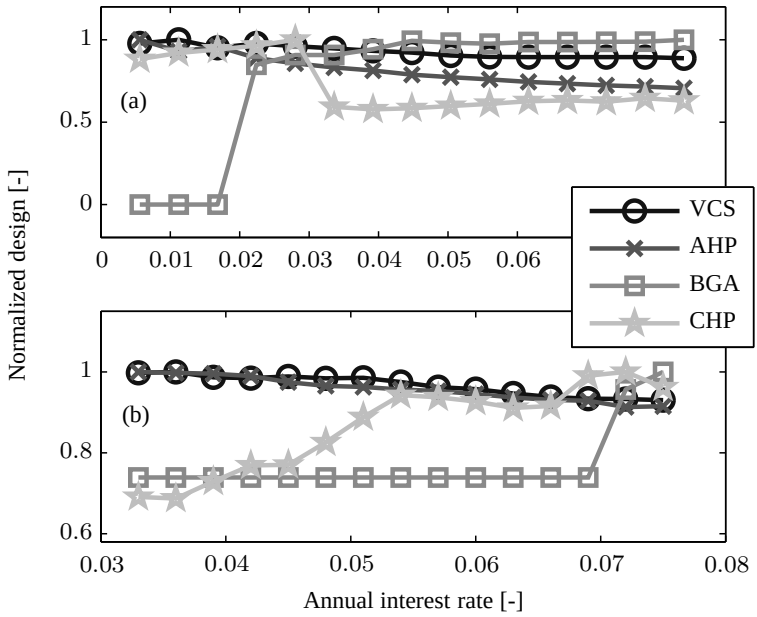


Figure 4.5: Deterministic designs for different values of interest rates. (a) Energy price is 0.09 €/kWh; (b) Energy price is 0.25 €/kWh.

price and the gas price:

$$P_{\text{energy}} = \frac{\overline{P}_{\text{el}} + P_{\text{gas}}}{2}. \tag{4.3}$$

Figure 4.5 shows the sensitivity of the design with respect to these parameters. The results suggest that the sensitivity to the annual interest rate is also dependent on the energy price. This phenomenon is seen by comparing Fig. 4.5a with 4.5b, where a shift of the switching point for the selection of BGA is observed. For a fixed energy price of 0.09 €/kWh (top figure), the switching point occurs close to the point with an interest rate of 2.8%, while for an energy price of 0.25 €/kWh (bottom figure) it happens at around an interest rate of 7%. The explanation for this difference is the fact that a higher energy price results in higher operating costs hence causing less restricted capital costs, which

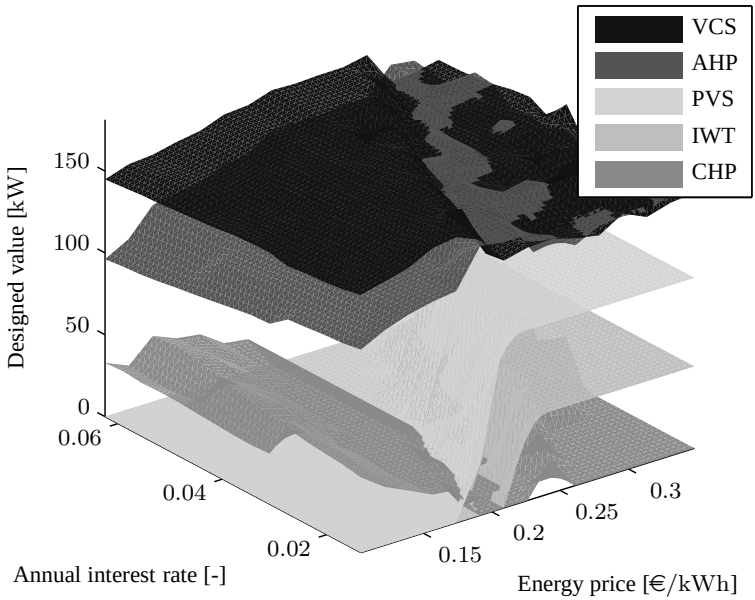


Figure 4.6: Deterministic designs for different interest rates and energy prices.

are bound to the interest rate. Due to this dependency, the sensitivity shows a nonlinear behavior.

In order to provide a better understanding of the sensitivity of the design variables, further deterministic optimizations have been carried out for various combinations of the annual interest rates and energy prices. The resulting designs are shown in Fig. 4.6. In the selected ranges of boundary conditions, the BGA is not selected and therefore not visible. Instead, in the optimal design, two renewable sources are included in the building energy system. The figure is particularly useful to identify the significant change in the selection and sizing of devices.

An important observation is the fact that in a certain range of the boundary conditions, the selection of several devices changes. This nonlinear behavior results in a highly sophisticated solution. In addition, the high sensitivity to the parameter changes implies a necessity to consider the uncertainties in the optimization process. A stochastic optimization is necessary, which results in a robustly designed building energy system.

4.5.3 Summary

The investigations on the scenarios discussed suggest that if uncertainties in boundary conditions are not neglected, the deterministic optimization provides a final optimal design only if the changes are smooth and the behavior is almost linear. In this case, the device sizes are chosen consecutively to guarantee comfort for the occupants for all probable scenarios. The sensitivity of the total system costs is rather small in this case, implying that pursuing a more sophisticated robust design only slightly improves the performance. On the contrary, if the sensitivity of the design shows a nonlinear behavior or if significant changes appear in the resulting device dimensions for different scenarios, the alternative approach of the robust design is clearly advantageous. However, the drawback of a robust design lies in its computational effort. The next section discusses the idea of utilizing a sensitivity analysis as a measure to determine whether a robust design is required or not. For these investigations, the uncertainties in the energy prices and the interest rates as well as the weather condition are considered.

4.5.4 Robust Design Approach

Figure 4.7 shows a two-dimensional projection of Fig. 4.6 on the X-Y plane, where the two characteristic regions *Alpha* and *Bravo* are defined. The high concentration of diagonal lines passing through the region *Alpha* is an indication of the high sensitivity to energy prices and annual interest rates, while the almost complete absence of such level curves in the region *Bravo* means a low design sensitivity. The goal is to compare the performances of a deterministic and a robust design, regarding the selection and dimensioning of devices, the total life-time costs and the computational time. Each of the regions *Alpha* and *Bravo* are sampled in multiple points in order to create several setups, as shown in Table 4.3. It is important to distinguish between a setup and a scenario. A scenario is a combination of uncertain values for the selected boundary conditions. In other words, it represents a sample from the uncertainty distribution. A setup defines which scenarios are used in the optimization. The setup 0 corresponds to a deterministic design with

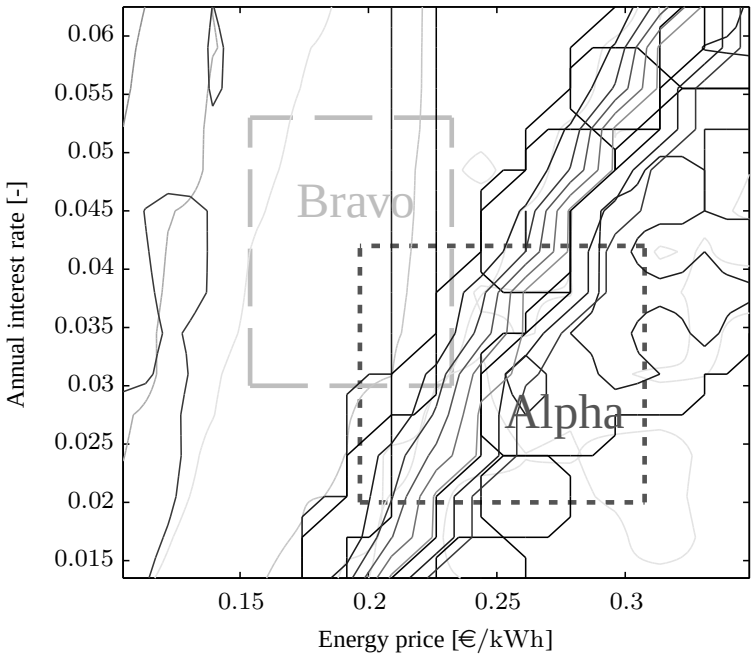


Figure 4.7: Projected level curves of Fig. 4.6: The sensitivity of design with respect to the two boundary conditions. The dash-marked regions *Alpha* and *Bravo* represent the high and low-sensitivity ranges, respectively.

the boundary condition values at the center of each region. Setups 1 and 2 expand the number of values included for the energy price and the interest rate, respectively. These are robust designs, although one of the boundary conditions is fixed so that its independent influence on the design can be examined. Setups 3 and 4 represent completely robust designs, with the difference that setup 4 also considers both normal and extreme weather profiles. Figure 4.8 shows the described setups on the level-curve plane. The resulting designs are summarized in Table 4.4 for region *Alpha* and in Table 4.5 for region *Bravo*. The tables show the optimal sizes for selected devices of each setup, as well as a normalized value of the total cost function.

Due to the change of several boundary conditions which affect the design, an explanation of the behavior is not straight-forward. Therefore,

Table 4.3: Setups in *Alpha* and *Bravo* regions. A maximum of 5 pricing scenarios and 5 interest rate scenarios are considered. Norm: normal weather; Norm/Extr: normal and extreme weather.

	Alpha		Bravo			
Setup	P_{energy}	r	P_{energy}	r	Weather	n_{sc}
0	0.25 (fixed)	0.031 (fixed)	0.20 (fixed)	0.042 (fixed)	Norm	1
1	[0.20...0.31]	0.031 (fixed)	[0.16...0.22]	0.042 (fixed)	Norm	5
2	0.25 (fixed)	[0.02...0.04]	0.19 (fixed)	[0.03...0.05]	Norm	5
3	[0.20...0.31]	[0.02...0.04]	[0.16...0.22]	[0.03...0.05]	Norm	25
4	[0.20...0.31]	[0.02...0.04]	[0.16...0.22]	[0.03...0.05]	Norm/Extr	50

the following general claims are to assist in clarifying the behavior of the optimizer:

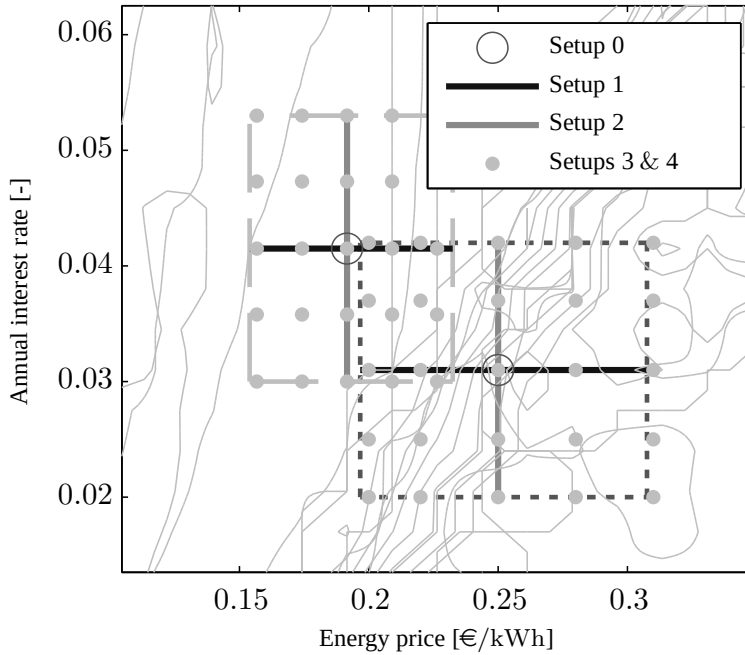


Figure 4.8: The setups listed in Table 4.3 shown on the level-curve plane.

Table 4.4: Optimal design (kW) for setups 0-4 in region *Alpha*.

Setup	PVS	IWT	AHP	CHP	VCS
0	**401**	**0**	**131**	**25**	**143**
1	800	40	129	46	143
2	372	0	131	23	143
3	523	30	131	34	143
4 (% change)	**800** (99%)	**26** (–)	**142** (8%)	**43** (72%)	**152** (6%)

Table 4.5: Optimal design (kW) for setups 0-4 in region *Bravo*.

Setup	PVS	IWT	AHP	CHP	VCS
0	**0**	**0**	**120**	**43**	**158**
1	0	0	120	42	159
2	0	0	120	43	158
3	0	0	120	43	158
4 (% change)	**0** (0%)	**0** (0%)	**133** (11%)	**35** (-19%)	**169** (7%)

1. Higher energy prices create a tendency towards installing more renewable sources, hence the building becomes more self-sufficient with the increased local production of electricity.

2. Higher interest rates encourage less investments, which means installing cheaper devices, therefore reducing the capital costs and at the same time increasing the operating costs.

3. Extreme weather conditions result in the over-dimensioning of almost every device in order to handle the peak power demands. If peak-electricity costs are considered, local electricity production is encouraged.

In the region *Alpha*, significant changes of the design are noticeable. The setup 0 is connected to a rather high energy price and a fairly low interest rate, therefore both PVS (photovoltaic system) and CHP are selected according to the first and second claims. The risk of higher energy prices in the robust design of setup 1 leads to a selection of both

renewable energy sources available. Also the CHP is over-dimensioned to increase the local electricity production. In contrast, the robust design of setup 2 considers a more variable interest rate, therefore the expensive renewable sources and co-generation unit are either deselected or under-dimensioned. The similarity of the resulting designs in setups 2 and 3 suggests that the risk of a high interest rate is more decisive in the design. However, when both normal and extreme weather conditions are included in the uncertainty list of setup 4, the fully robust design includes high-dimensioned local producers, due to the third claim stated above. Table 4.4 also shows the percentage of changes in setup 4 with respect to setup 0. The numbers suggest a substantial difference in the final optimal design for the two cases of deterministic and stochastic optimizations. Therefore, if the area of the interest for the boundary conditions is the region *Alpha*, a robust design is advantageous.

In the region *Bravo*, on the other hand, the situation is different. Table 4.5 shows the optimal design for the similar setups. The small differences in both selection and sizing implies a low sensitivity of the design to the changing boundary conditions. Even comparing the fully robust setup 4 with the deterministic setup 0 reveals the fact that a robust design is only slightly advantageous in this case. When the high computational effort of a robust design is taken into account as well, the conclusion is that for the ranges of uncertain parameters in region *Bravo*, a deterministic design is sufficient.

In order to make a more clear comparison between the deterministic and robust designs for each region, the performance of such approaches are investigated for all the 50 scenarios, as shown in Table 4.6. In both regions, the robust design demands for higher investigations on the devices installed, as denoted by O^{inv}. However, the total annual cost of all possible scenarios is reduced by almost 483,000 Euros in the region *Alpha*, and almost 75,000 Euros in the region *Bravo*. The results suggest that the usage of a more computationally expensive robust design pays off due to a consideration of uncertainties. However, the number of uncertainty scenarios in the optimization process is extensively limited by the computational power.

Table 4.6: Performance comparison for deterministic and robust designs in the region *Alpha* (top) and *Bravo* (bottom). Costs are in Euros and are calculated for one year and for all scenarios.

Region *Alpha*

	O^{inv} [€]	O^{dcm} [€]	O^{opr} [€]	O^{peak} [€]	Total [€]
Deterministic Design	475 207	113 825	6 714 899	648 883	7 952 814
Robust Design (% change)	856 713 (80%)	107 295 (-6%)	5 882 993 (-12%)	622 630 (-4%)	7 469 631 (-6%)

Region *Bravo*

	O^{inv} [€]	O^{dcm} [€]	O^{opr} [€]	O^{peak} [€]	Total [€]
Deterministic Design	182 511	111 783	5 718 404	680 188	6 692 886
Robust Design (% change)	193 465 (6%)	116 391 (4%)	5 630 508 (-2%)	677 456 (-0.4%)	6 617 819 (-1%)

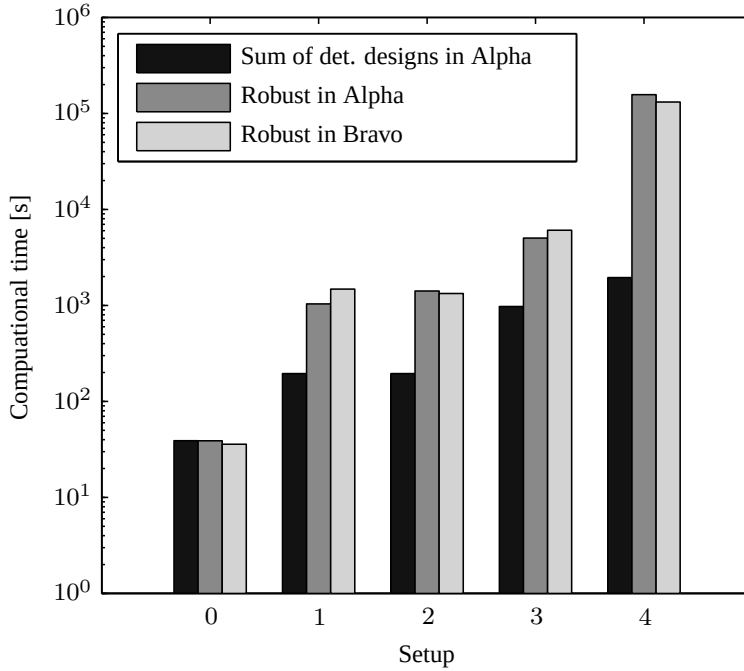


Figure 4.9: Computational time for different setups in regions Alpha and Bravo. A black bar shows the total computational time required to pursue a deterministic design for each scenario.

4.5.5 Discussion

Figure 4.9 shows the computational time for various setups in both regions. The calculations are carried out on an Intel Xeon E5-2430 with 6 cores clocking at 2.70 GHz. Obviously, the solving time increases exponentially when more uncertainties are considered. In particular, for the setup 4 a robust design takes approximately 100 times longer than a summation of deterministic designs for each individual scenario. This fact again proves that a sensitivity analysis is a useful assessment method to examine the necessity of a robust design.

Remarkably, solving the robust problem in just one step guarantees a globally optimal solution, whereas solving n deterministic problems generates n local cost minima. In the latter case, several methods such

as a two-step Monte Carlo approach can be used to find a single final optimal design. One advantage of a two-step approach would be a resulting linear rather than exponential increase of computational time as a function of the number of scenarios in the optimization.

5 Conclusions

5.1 Summary

The potential for the optimization of building services is increased due to the applications of renewable energy sources and storage technologies. Many researchers have addressed the issues regarding the design and control of building systems in the recent years, resulting in methods and tools which enable a long-term simulation of thermal and electrical behavior of buildings be carried out within minutes. However, with new energy conversion and storage techniques being introduced, there is a need for general modular frameworks to facilitate the process of total building optimization.

The focus of this thesis is on the development of such a modular framework, namely the Smart Building Designer, which enables a simultaneous optimal design and control of a building, as well as sensitivity analyses over local or external parameters. The framework can be used not only by academic researchers, but also by lawmakers to investigate the effect of energy and environmental directives on the long-term behavior of residential and commercial buildings. Arbitrary devices and energy carriers can be added by the user and cumulative energy demand and life cycle carbon footprint can be assessed using the framework. In order to prove the potentials of the framework, several case studies are accomplished in this thesis.

In the first case study, it was shown that the framework is able to solve a deterministic optimization of the total building system for one year in less than 10 minutes on a typical desktop computer. This ability is mainly due to the accurate but control-oriented formulation of mixed integer-linear models. The number of independent boolean variables are limited to the number of devices included, in order to speed up

the optimization process even further. The framework pre-calculates the required boundary conditions using the raw data such as those gained from meteorological measurements (temperature and solar irradiation), occupancy schedule, and spot-market electricity rates. As a result, the maximum flexibility is provided, enabling the user of SBD to apply the framework to an arbitrary building in any location. The sensitivity analyses also demonstrate how increasing energy-carrier prices or governmental legislation can result in different optimal designs. Most interestingly, the results suggest that an installation of renewable and storage devices is only beneficial when strict limitations are applied on the consumption of energy, while an increase of the electricity prices does not create the same effect on the selection of devices.

In the second deterministic design case study, a comprehensive investigation of the optimization of the cooling system under various electricity tariffs is performed. The cost saving potential of ice storage system and its advantage in helping the cooling system respect the comfort constraints of the building temperature is successfully shown. By adding an ice storage system, the total costs are reduced by 11% under the two-level electricity tariff scheme, while the cooling power capacity of the vapor compression system is lowered by 26%. The analysis of the optimal cooling system operation reveals that a cost minimization is achieved by equalizing the cooling power production using the ice storage during peak cooling loads.

Based on the spot market data, grid loading factors, taxes, and power regulation market data, a variable electricity tariff is developed. Including an ice storage system, cost reductions of 5.7% are achieved under this tariff scheme. The analysis further reveals that cooling systems make optimal use of the power regulation market by producing a considerable amount of the cooling energy during times of low or negative electricity prices. However, the main target of including ice storage systems is to equalize the cooling power production, to reduce the peak power consumption, and to downsize the vapor compression system.

Finally, the effects of uncertainty in the boundary conditions are discussed in the robust design case study. The framework is extended to perform stochastic optimizations. The pre-generated uncertainties create a set of so-called scenarios, which are translated into the correspond-

ing objective functions. Each scenario has an individual optimal operation strategy, but the optimizer returns a single optimal result for the selection and the dimensioning of devices regardless of the number of scenarios, known as a robust design. The drawback of a robust design is its high computational time that increases exponentially by adding to the number of scenarios in question. In order to avoid an unnecessarily sophisticated optimization of a complex multi-scenario problem, a sensitivity analysis is used to assess the fragility of the solution with respect to changes in boundary conditions. If the sensitivity is low-valued and smooth, a computationally efficient deterministic optimization yields a performance similar to that of the robust one. In the case of a high sensitivity or a nonlinear behavior, a robust design approach must be used over a set of well-chosen ranges of the boundary conditions. An intensive analysis is conducted to investigate the necessity of a robust design in the case study, showing that a significant benefit only appears on a certain range for the case presented. Additionally, a visual representation of the sensitivity analysis assists in choosing a proper range of boundary conditions, especially when more than one of them is subject to uncertainties.

5.2 Outlook

The research carried out under the scope this thesis can be continued in several directions, mainly to expand the developed framework. Some ideas are described here.

- The current framework inhabits the optimal control as the main type of control system. Since this approach requires large computational power, it could be interesting to compare the results to the case with a simpler operating strategy, such as a rule-based control. The performance of such method can be similar, while it is simpler to implement it on a real system.

- A further development of the framework can be a risk-aware optimization approach, which minimizes not only the expected values

of the overall system costs, but also other robustness criteria such as the risk of high costs.

- The interaction with the smart grid can be investigated in more details. The building services can be optimized to handle bidirectional connections such as demand-response and peak-shaving programs in a more efficient manner.

- An extension can be to target a multiple buildings problem. The buildings will be allowed to exchange different types of energy such as electricity, heat and hot water. Ultimately, the framework can be used for optimization of energy flows within blocks, city zones or even cities.

Appendix A

Nomenclature

Latin Symbols

A	area	[m^2]
C	capacity	[J/kgK] or [J]
D	dimension	[unit][1]
E, $\mathbf{E}$	electric energy	[kWh]
$\dot{E}$, $\dot{\mathbf{E}}$	electric power	[W]
f	factor	[unit]
H, $\mathbf{H}$	chemical energy	[kWh]
$\dot{H}$, $\dot{\mathbf{H}}$	chemical power	[W]
$\mathbf{I}$	solar irradiation	[W]
l	length	[m]
L	legislation	[unit]
$load$	load level (demand) in the grid	[W]
n	number	[–]
N	optimization duration	[time steps]
O	objective term	[€]
P, $\mathbf{P}$	price or cost	[€/unit]
p	probability	[–]
Pa	pressure	[Pa]

[1] The variable may obtain different units according to the application.

Q, $\mathbf{Q}$	thermal heating energy	[kWh]
$\dot{Q}$, $\dot{\mathbf{Q}}$	thermal heating power	[W]
Q', $\mathbf{Q}'$	thermal cooling energy	[kWh]
$\dot{Q}'$, $\dot{\mathbf{Q}}'$	thermal cooling power	[W]
r	interest rate	[–]
R	thermal resistance	[K/W]
S	subsidy	[€] or [€/unit]
T, $\mathbf{T}$	temperature	[K]
t_s	time step	[s]
U	thermal transmittance	[W/K] or [W/m^{2}K]
v	velocity	[m/s]
V	volume	[m^3]
w	width	[m]

Greek Symbols

χ	existence variable	[–]
η	efficiency	[–]
γ	power coefficient	[1/s]
ρ	density	[kg/m^3]
π	constant parameter	[unit]
σ	self-discharge rate	[–]
τ	time horizon or lifetime	[years]

Acronyms

ACS	absorption chiller system
AHP	air-source heat pump
BAT	battery
BDG	building

BEL	electric boiler
BGA	gas boiler
CHP	combined heat and power
CAS	compressed air storage
CON	converter devices
COP	coefficient of performance
CWS	cool water storage
EXT	external sources
FL	full-load operation
GHP	ground-source heat pump
ISS	ice storage system
IWT	integrated wind turbine
MILP	mixed integer linear programming
NPV	net present value
NS	non-standard condition
PL	part-load operation
PCM	phase changing material
SBD	smart building designer
STO	storage devices
TES	thermal energy storage
VCS	vapor compression system

Indices

amb	outdoor ambient air
ch	charging
CO_2	related to carbon dioxide
d	decision variable
dist	distribution
dch	discharging
dhw	domestic hot water
el	electricity
eqp	equipments

in	input/inlet/inflow
inv	investment
loss	related to energy loss
max	maximum
min	minimum
occ	occupants
opr	operational
out	output/outlet/outflow
prp	proportional
ref	reference value
reg	regulation
sc	scenario
set	set-point
sol	solar irradiation
spot	related to spot market
tot	total
ven	ventilation

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